# PHASE DIAGRAMS

***Some definitions…***

**Component:** element or chemical compound whose presence is necessary and sufficient to make a system.

**Alloy:** mixture of two or more elements having metallic properties.

**Binary systems → two components.**

**Phase:**

A phase is a **homogeneous portion of a system with uniform chemical and physical characteristics** (therefore, both in terms of structure and composition).
A single phase has the same behavior in all its parts when subjected to physical or chemical stresses

A phase can consist of more than one chemical component (such as water or metallic solid solutions in which some atoms (solutes) are placed within the crystalline structure of te other (in interstitial or substitutional positions)

A chemical element or compound can exist as different phases (liquid water, water vapor and ice, Fe$\alpha$, Fe$\gamma$)

# MONOPHASIC SYSTEMS

## ONLY ONE PHASE

WATER liquid

ICE solid

WATER vapour

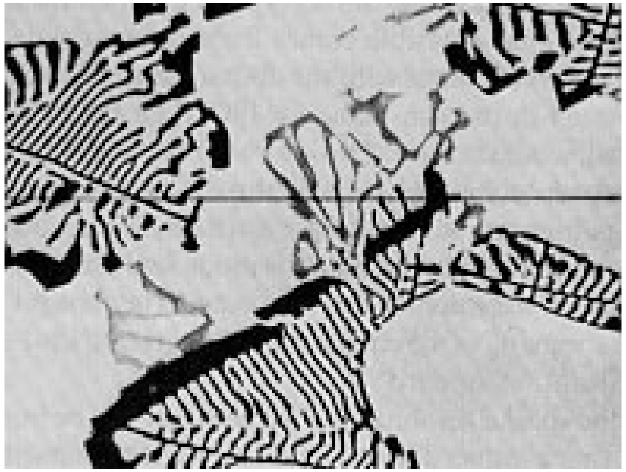
Fe FCC solid

OIL liquid

# BIPHASIC SYSTEMS

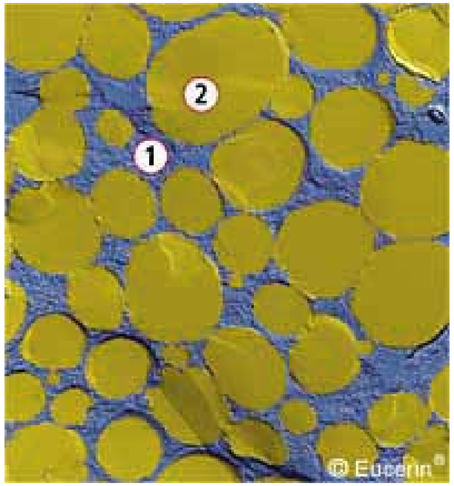
**They are made of TWO different phases that differ in terms of:**

- **Aggregation state (e,g. liquid vs. solid) and/or**
- **Composition and/or**
- **Crystalline structure**



Solid solution +
second phase

Water and
ice




Water
and oil

Properties of materials depend on microstructure → it depends on **composition, T, p**

Equilibrium diagrams or phase diagrams are graphical representations of the various phases (microstructures) in a system depending on **composition and temperature**

All the phase diagrams have:
- T on the y-axis
- composition (wt.%) on the x-axis

A phase diagram is:
a graph describing how many and which phases of a system are present when the intensive parameters (pressure, temperature) of the system itself vary.

Phase diagrams are valid **when the thermodynamic equilibrium is reached** (slow cooling).

Phase diagrams can be obtained experimentally or by calculations.

In 1-component phase diagrams, we consider both T and p.

In binary phase diagrams, we consider only the condensed phases (solid and liquid) → p is considered not significant (constant).

Phase diagrams are very useful for different applications; for example, thanks to them you can:

- Determine the number and type of phases as temperature and composition vary;

- Calculate the relative quantities of the phases present at the equilibrium;

- Determine the maximum solubility, in equilibrium conditions, of one component in another;

- Know the melting temperature of the various phases;

- Determine the temperature at which phase transformations occur.

## Gibbs' rule or phases rule

Gibbs' rule (phases rule) establishes mathematically the relationship between the number of variables that can change (V), the number of components (C), the number of phases (f) and the number of physical parameters (p, T):

$$\mathbf{V = C - f + n}$$

Since p = cost, then the Gibbs' rule can be rewritten as:

$$\mathbf{V = C - f + 1}$$

In binary systems C = 2, thus:

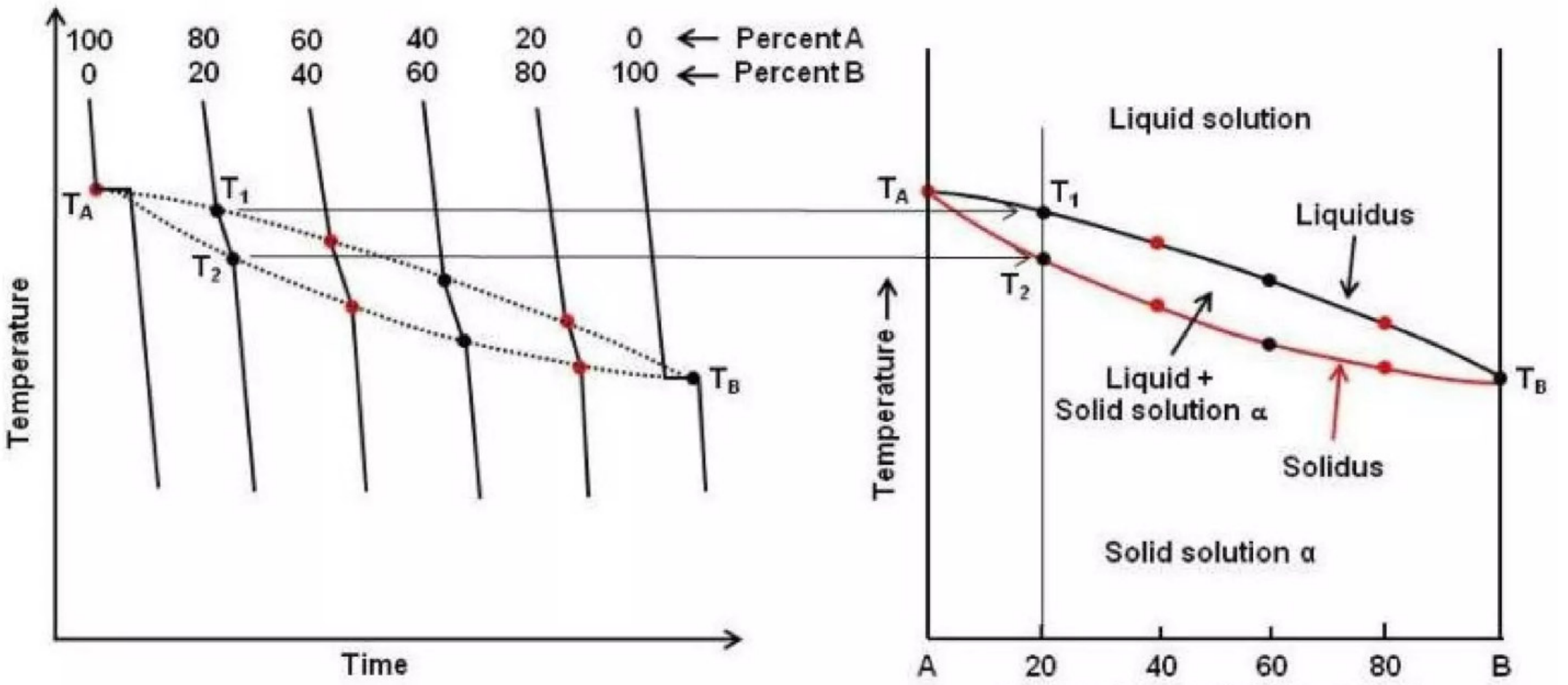
$$\mathbf{V = 3 - f}$$

# Cooling Curves

- cooling curve is the graphical plot of phases of element on temperature v/s time.
- The resulting phase during solidification is different for various alloy composition.
- The most common coolingcurves are:

1. For pure metals
2. For binary solid solution(alloy)
3. For eutectic binary alloy
4. For off-eutectic binary alloy

- Series of cooling curves:




Series of cooling curves for different alloys in a completely soluble system both in the liquid and solid states

Composition, weight percent B

Phase diagram of two metals completely soluble in the liquid and solid states

**Construction of Phase Diagram from Series of Cooling Curves**

# Classification of phase diagrams

- Full solubility in the solid state

- No solubility in the solid state (eutectic)

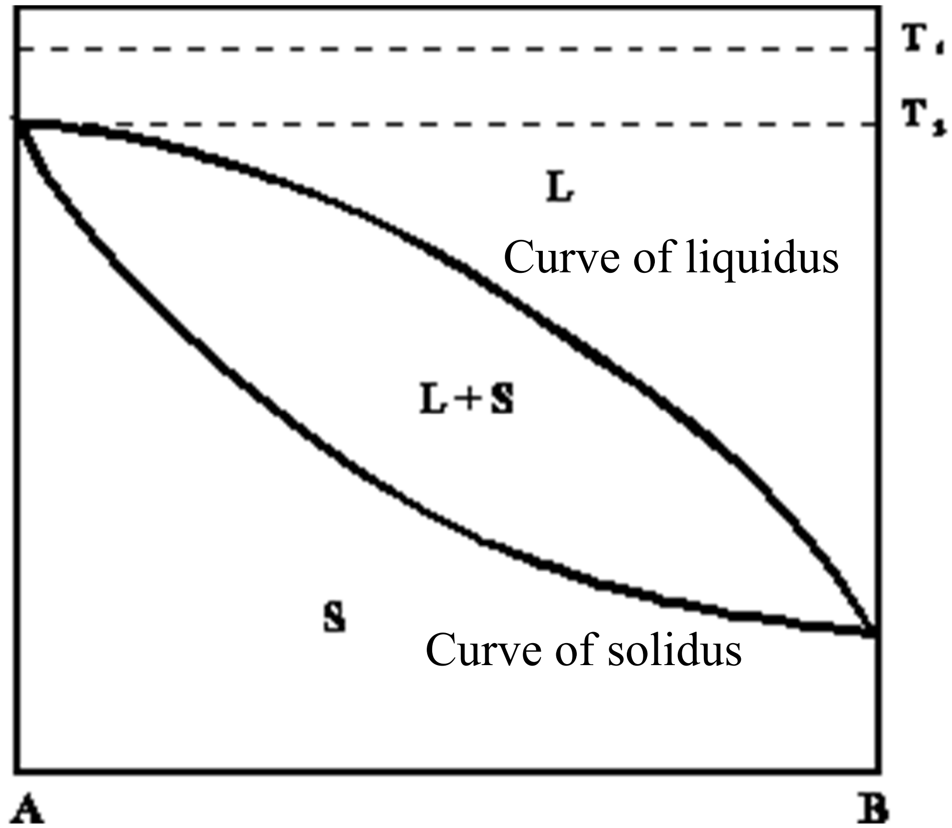
- Partial solubility in the solid state (eutectic)

- Peritectic

- Presence of congruent/non-congruent melting

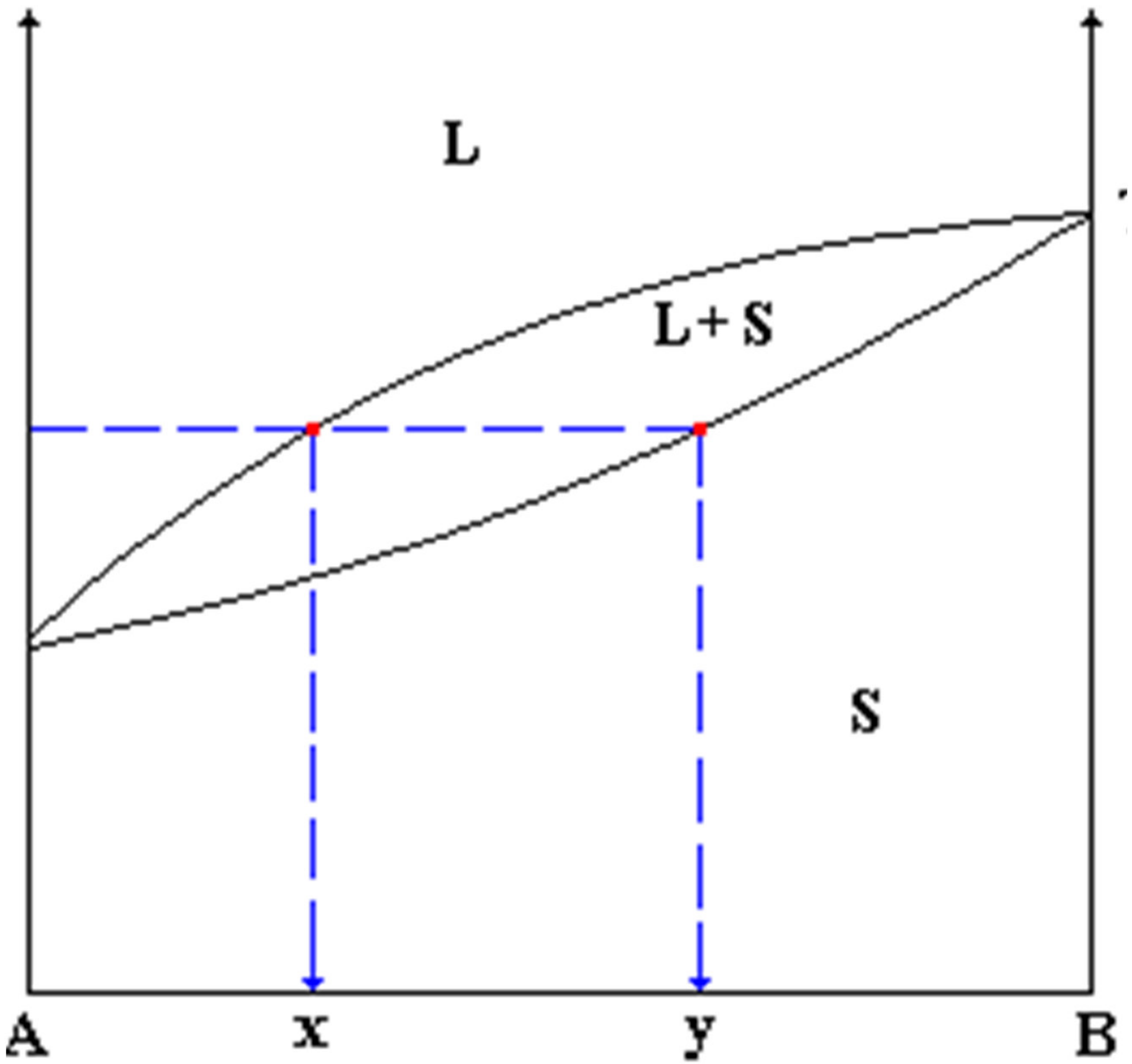
# Complete solubility in liquid and solid state

e.g. Cu-Ni or Ag-Au.



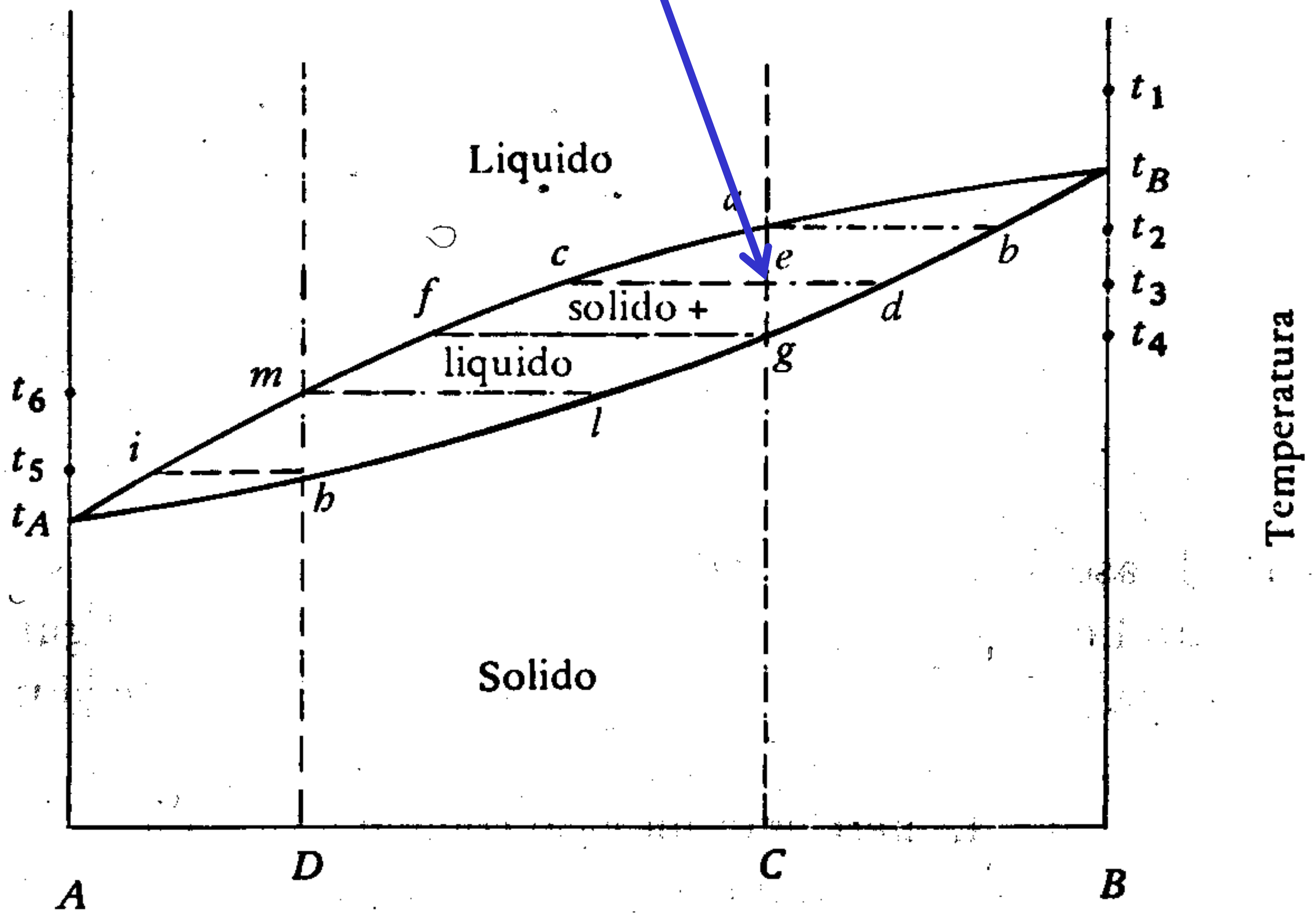

**V=C-f+n**

## How to read composition of phases



# How to read % of phases

$$\% \text{ liquido} = \frac{ed}{cd} 100 \qquad \% \text{ solido} = \frac{ce}{dc} 100$$



# How to read the structure morphology

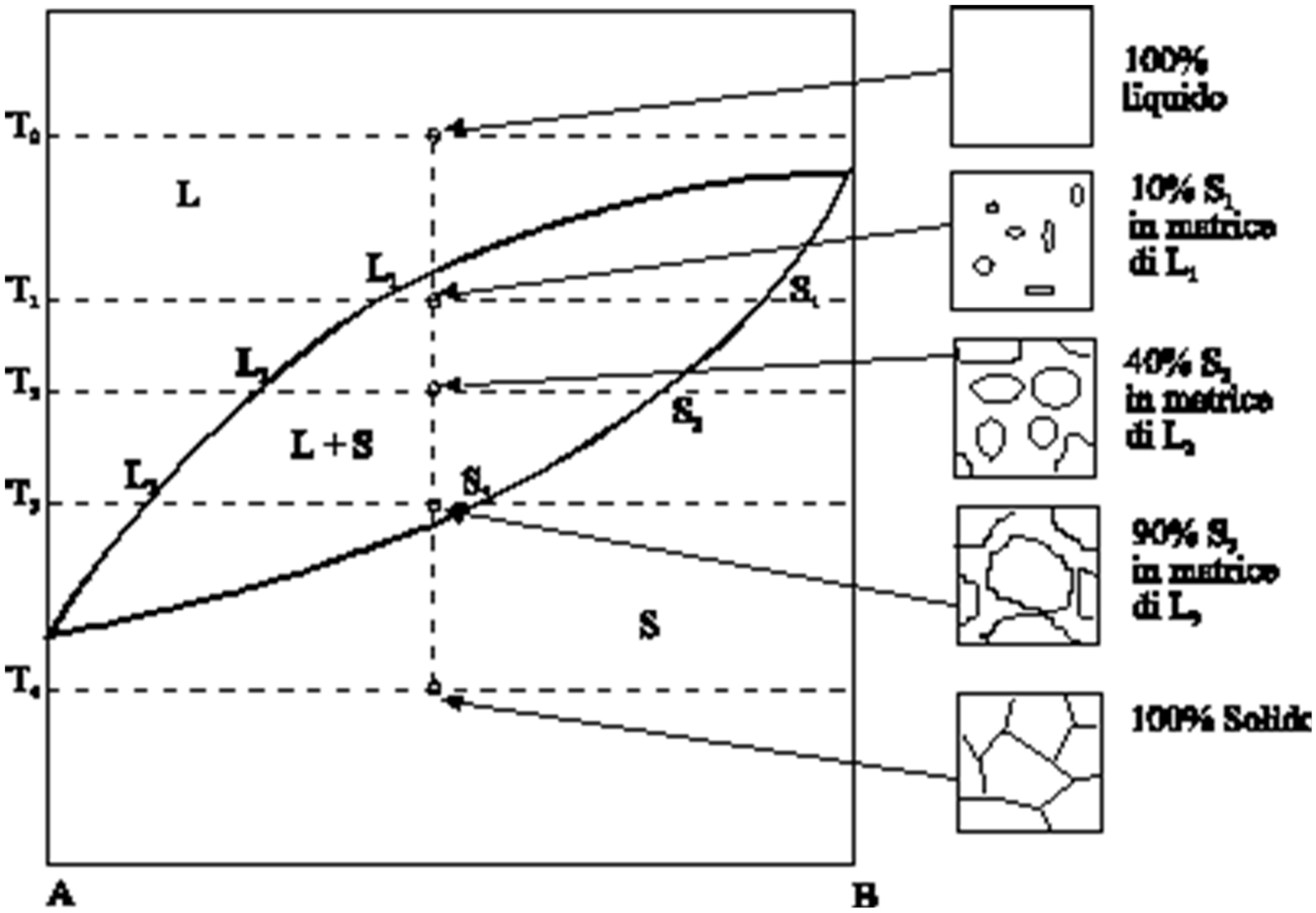
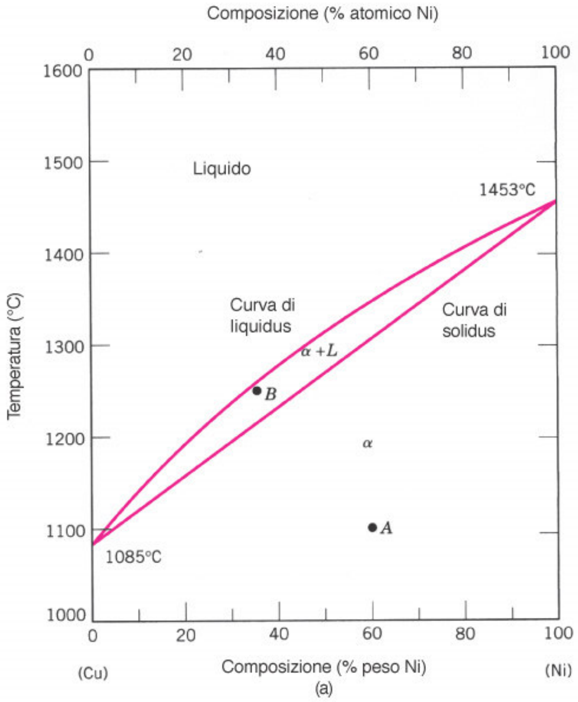


**FIGURA 9.2** (*a*) Diagramma di fase rame–nichel. (Da *Phase Diagrams of Binary Nickel Alloys*, P. Nash, Editor, 1991. Ristampa autorizzata da ASM International, Materials Park, OH.) (*b*) Parte del diagramma di fase rame–nichel in cui si osserva come al punto B siano determinate sia la composizione che le quantità delle fasi presenti.



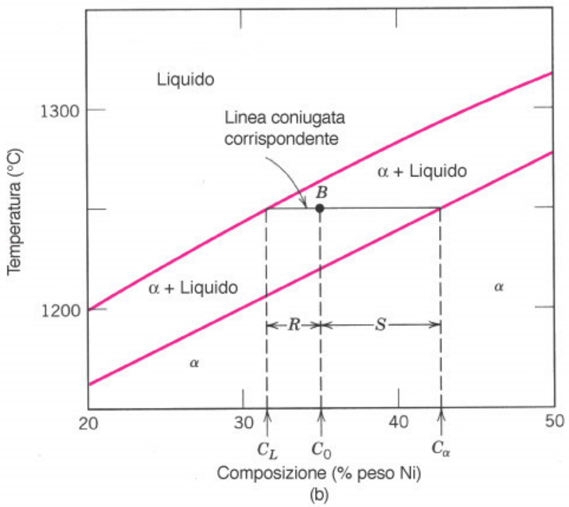
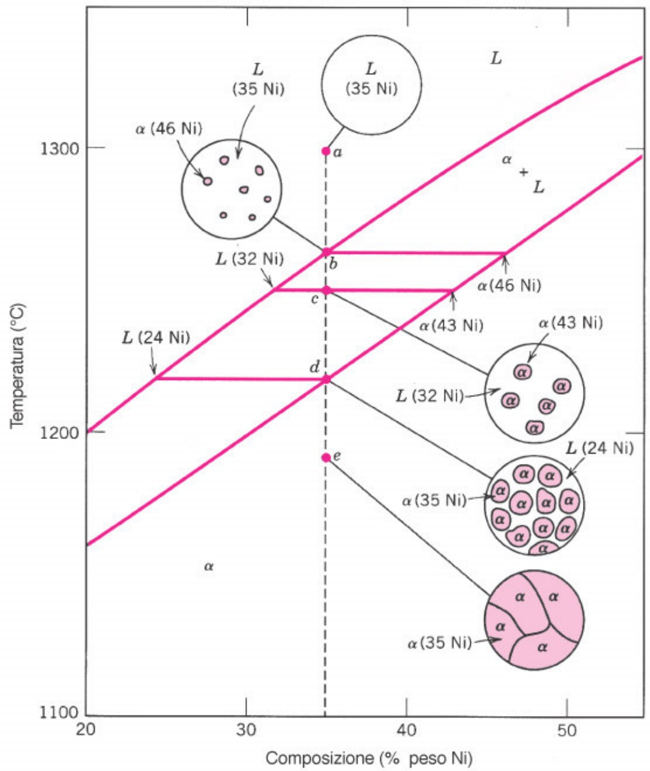




**FIGURA 9.3** Rappresentazione schematica dell'evoluzione della microstruttura durante la solidificazione di equilibrio di una lega con 35% in peso di Ni e 65% di Cu.

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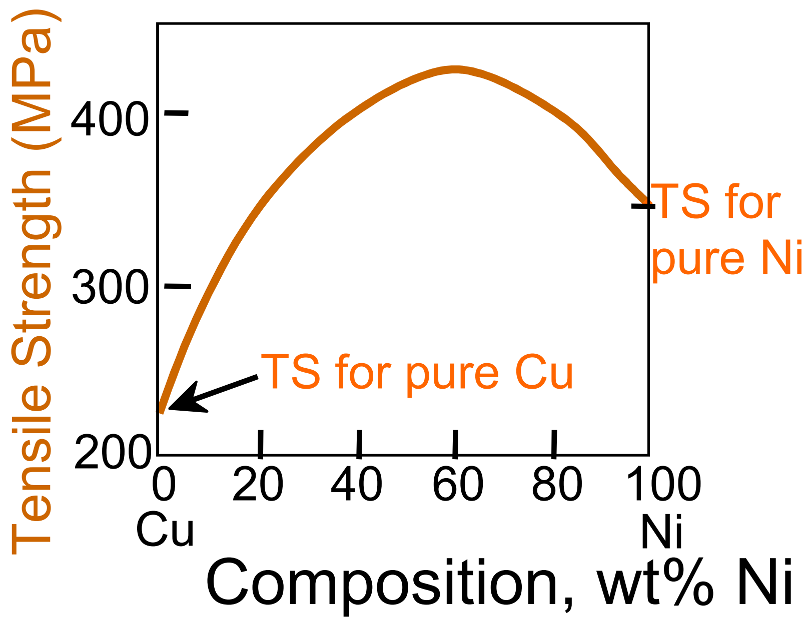
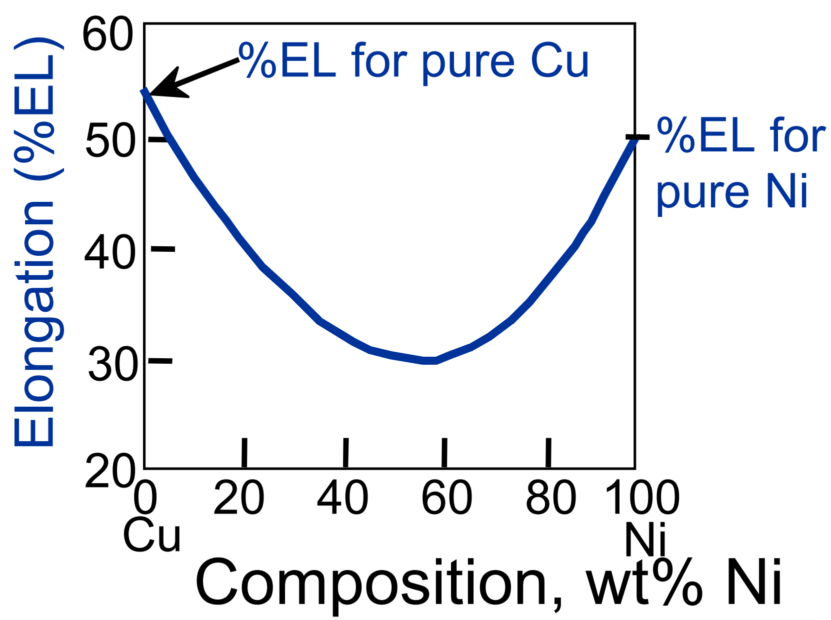
## Effect of solid solution strengthening on:

### Tensile strength (TS)




Adapted from Fig. 9.6(a), Callister & Rethwisch 8e.

### Ductility (%EL)



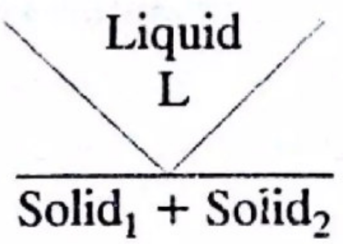

Adapted from Fig. 9.6(b), Callister & Rethwisch 8e.

- # EUTECTIC SYSTEM

"Eutectic" (from Greek) = "eu", which means something easy or good, and "tactic", which is the process of melting. So the general meaning in Greek is *something that melts in a good manner*.

  - In an eutectic reaction, when a liquid solution of fixed composition, solidifies at a constant temperature, forms a mixture of two or more solid phases without an intermediate pasty stage. This process reverses on heating.

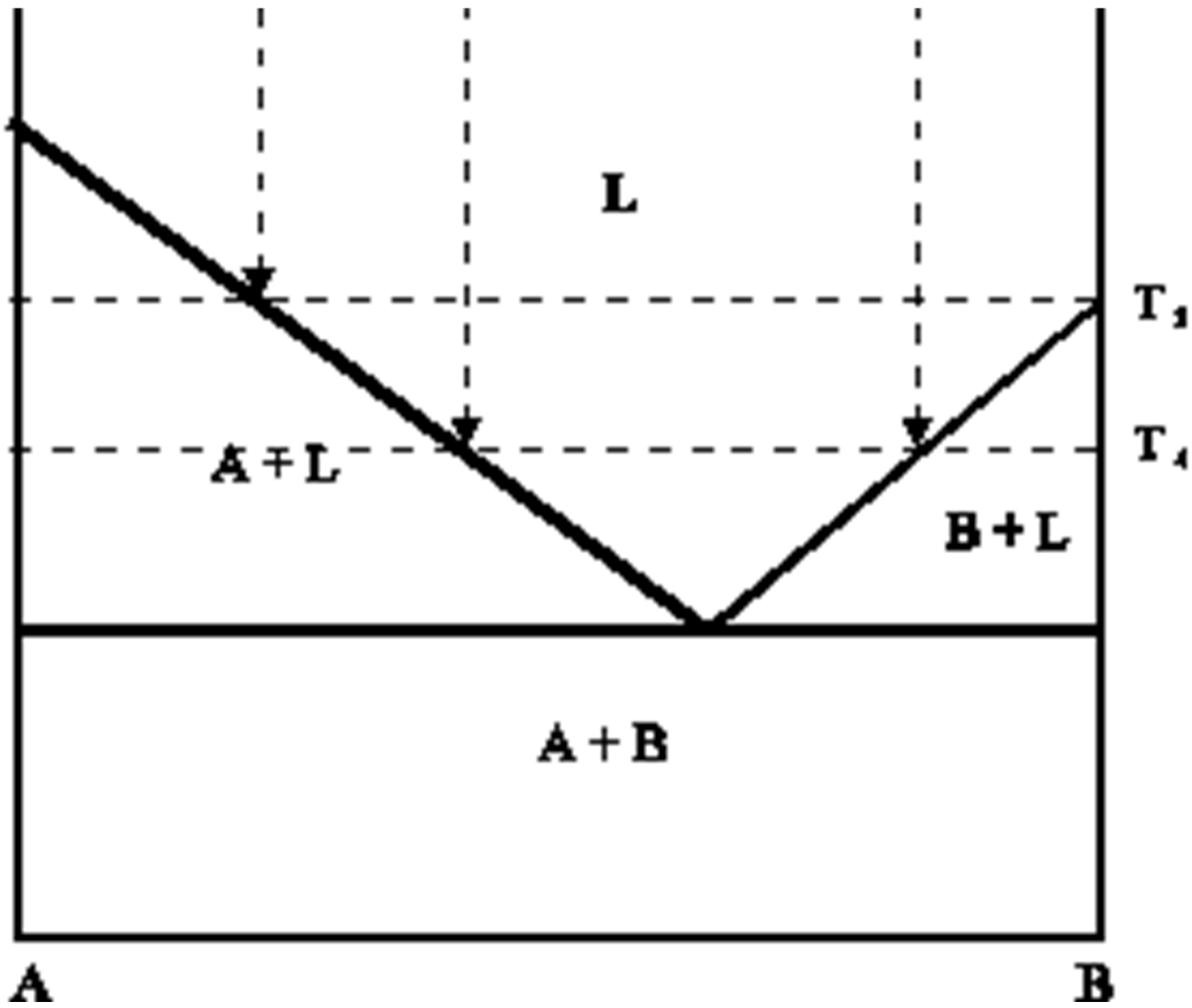
$$\text{Liquid} \underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}} \text{Solid}_1 + \text{Solid}_2$$



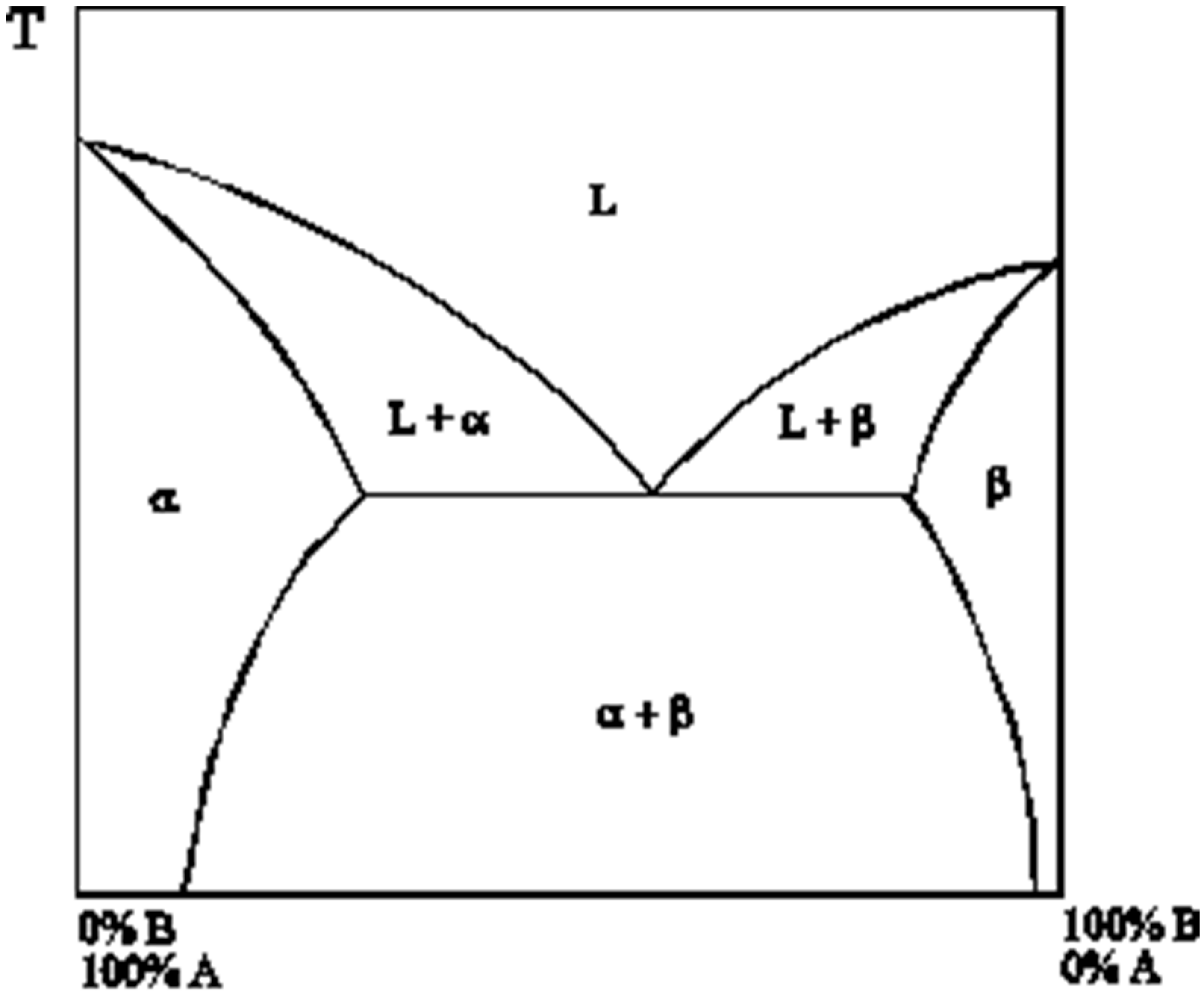

$$L \leftrightarrow A + B \quad \text{or} \quad L \leftrightarrow \alpha + \beta$$

  - In eutectic system, there is always a specific alloy, known as eutectic composition, that freezes at a lower temperature than all other compositions.
  - At the eutectic temperature, two solids form simultaneously from a single liquid phase.
  - The eutectic temperature and composition determine a point on the phase diagram called the eutectic point.

# Complete solubility in the liquid state, no solubility in the solid state



# Complete solubility in the liquid state, partial solubility in the solid state



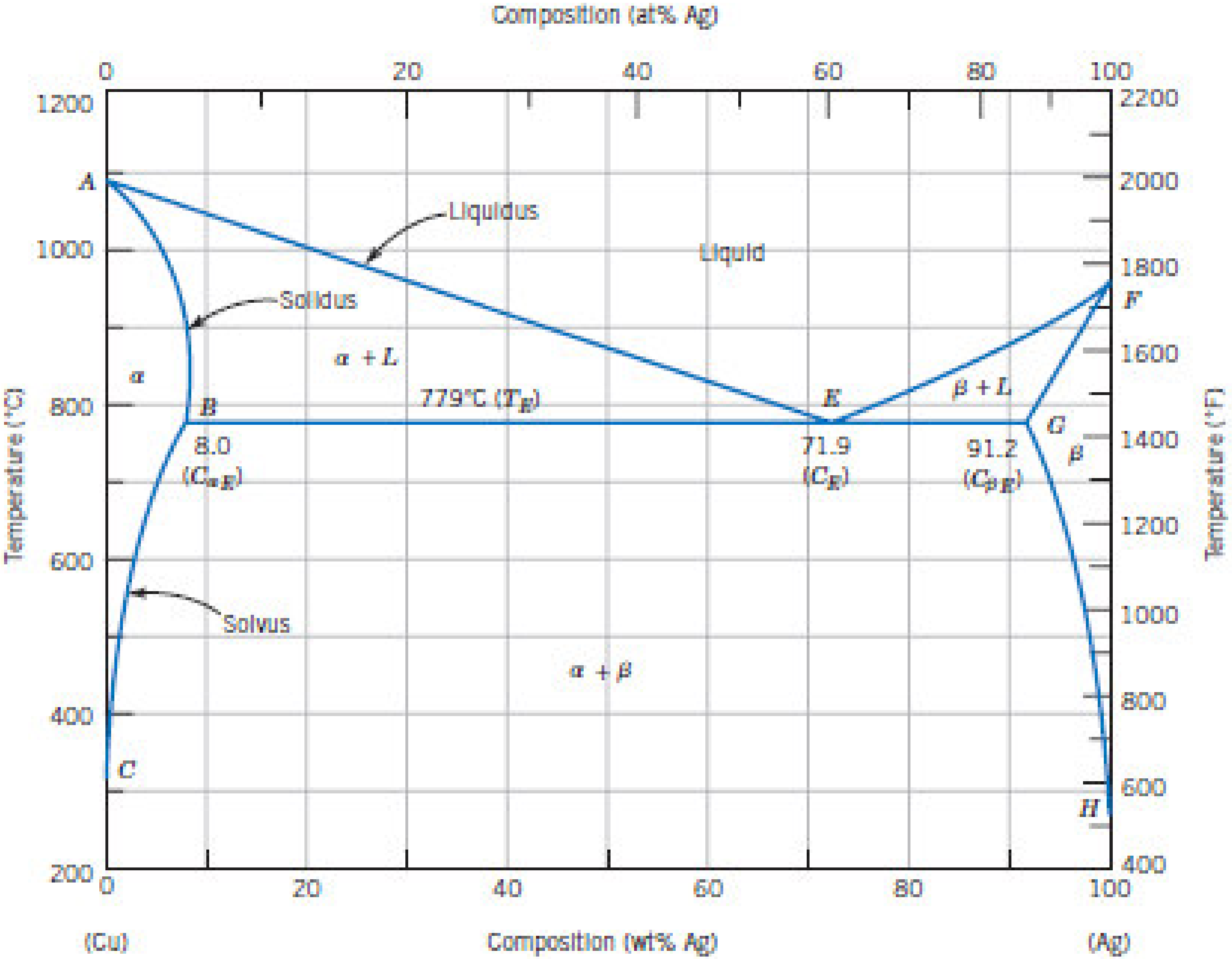
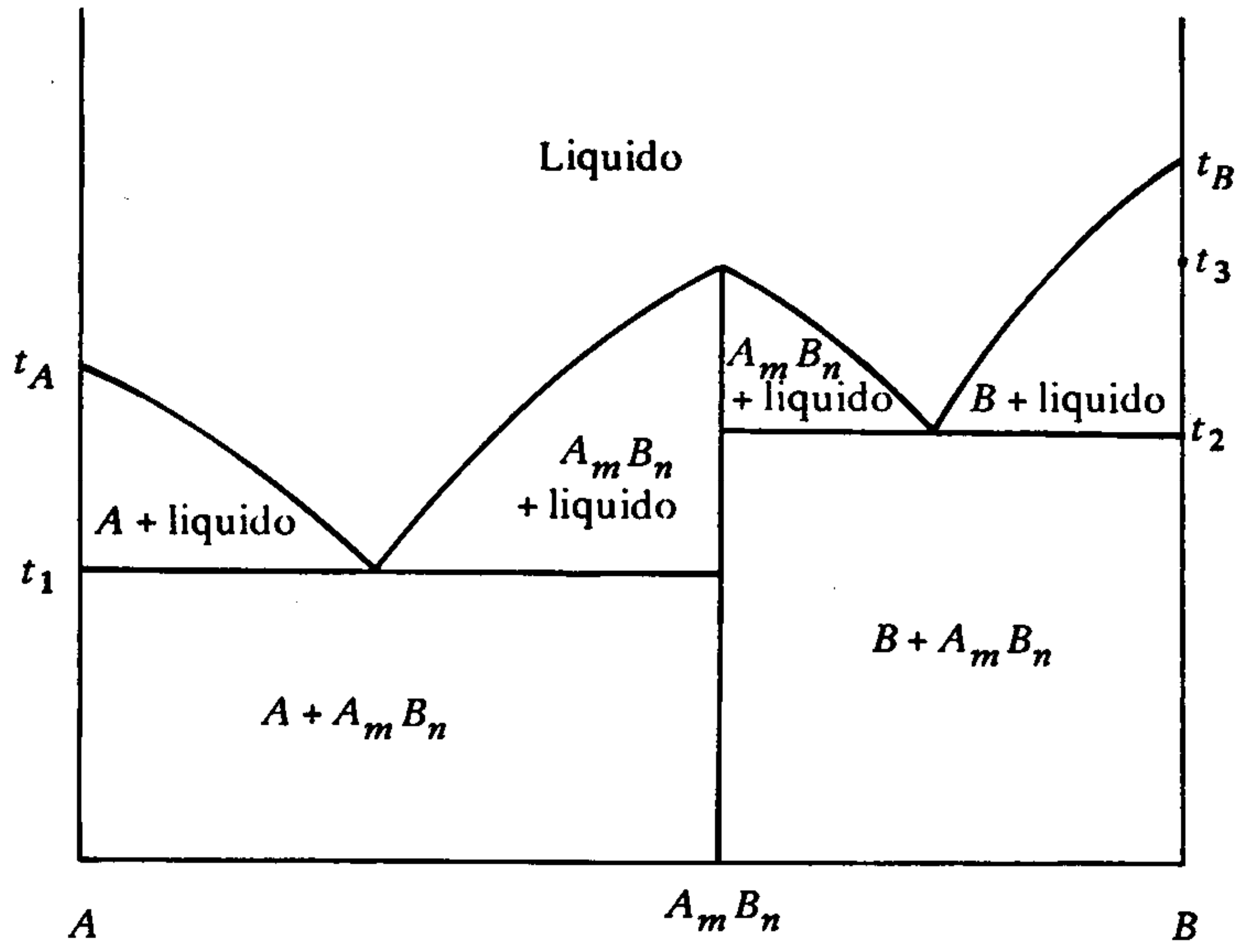



**Figure 9.7** The copper–silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

# Formation of a compound with congruent melting

In this case: complete solubility in the liquid state, no solubility in the solid state



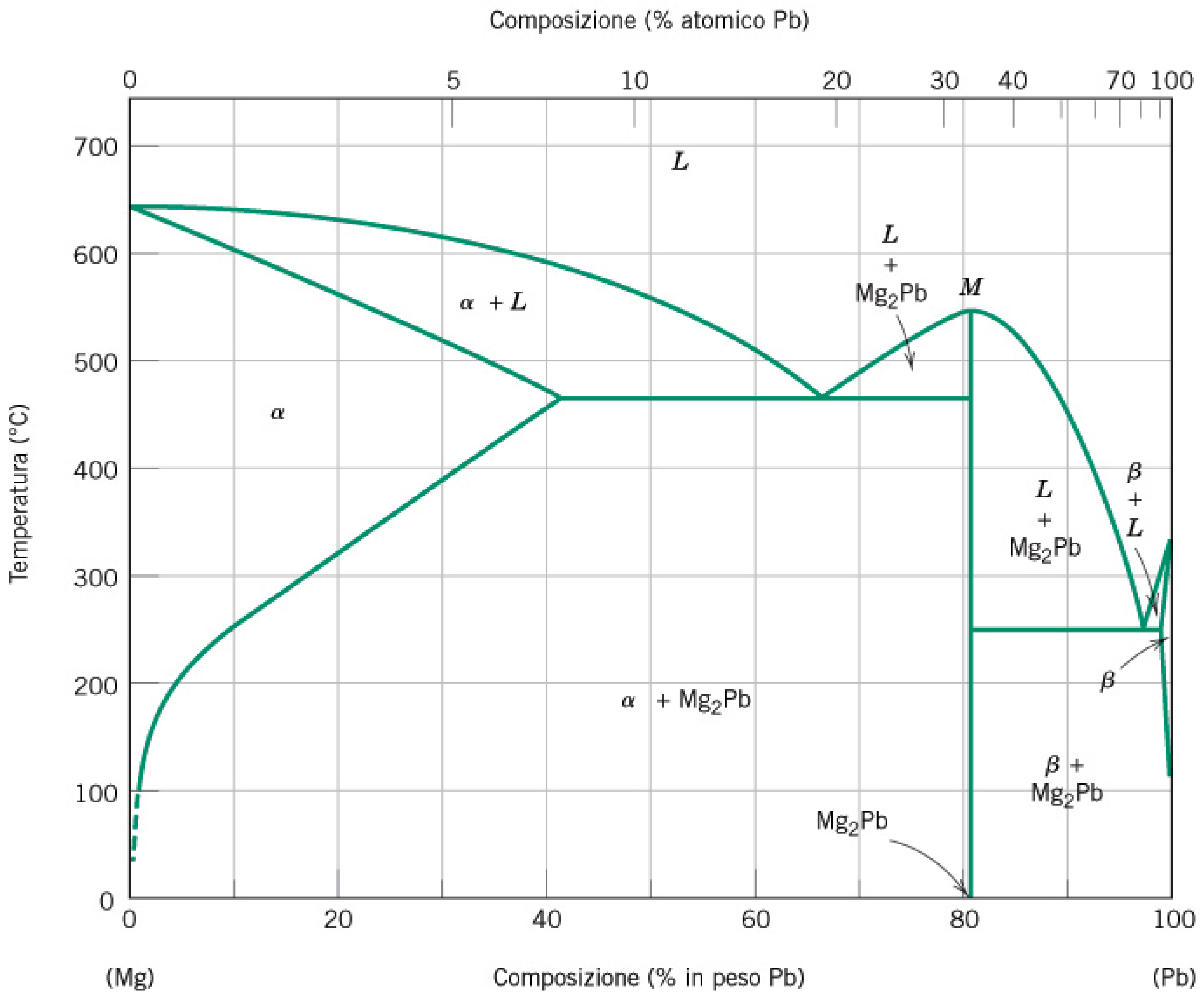
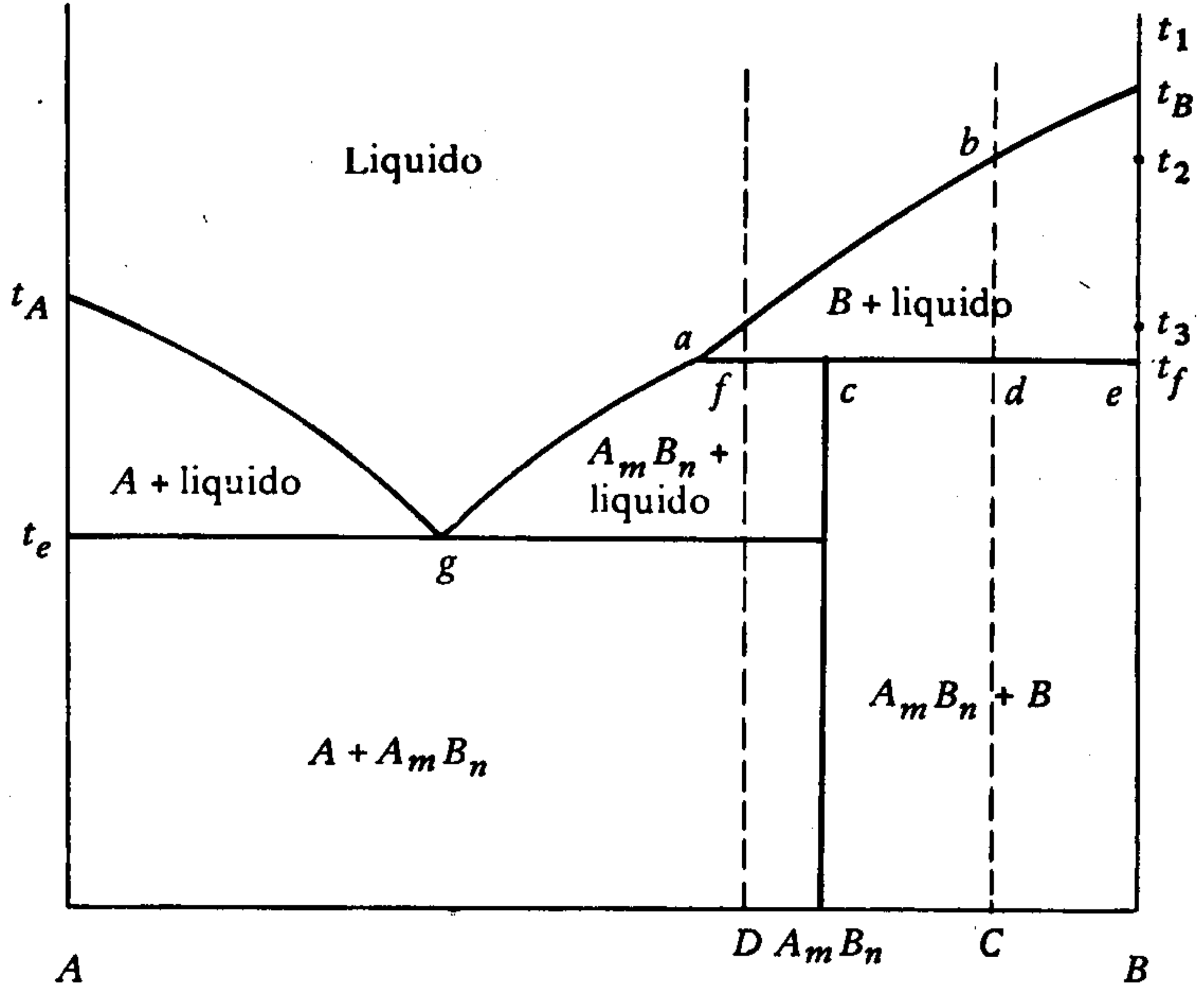



**Figura 9.20** Diagramma di fase magnesio–piombo. [Da *Phase Diagrams of Binary Magnesium Alloys*, A.A. Nayeb–Hashemi and J.B. Clark (Editors), 1988. Ristampa autorizzata da ASM International, Materials Park, OH].

# Formation of a compound with non-congruent melting

In this case: complete solubility in the liquid state, no solubility in the solid state



# Peritectic transformation

$L + \delta \leftrightarrow \gamma$

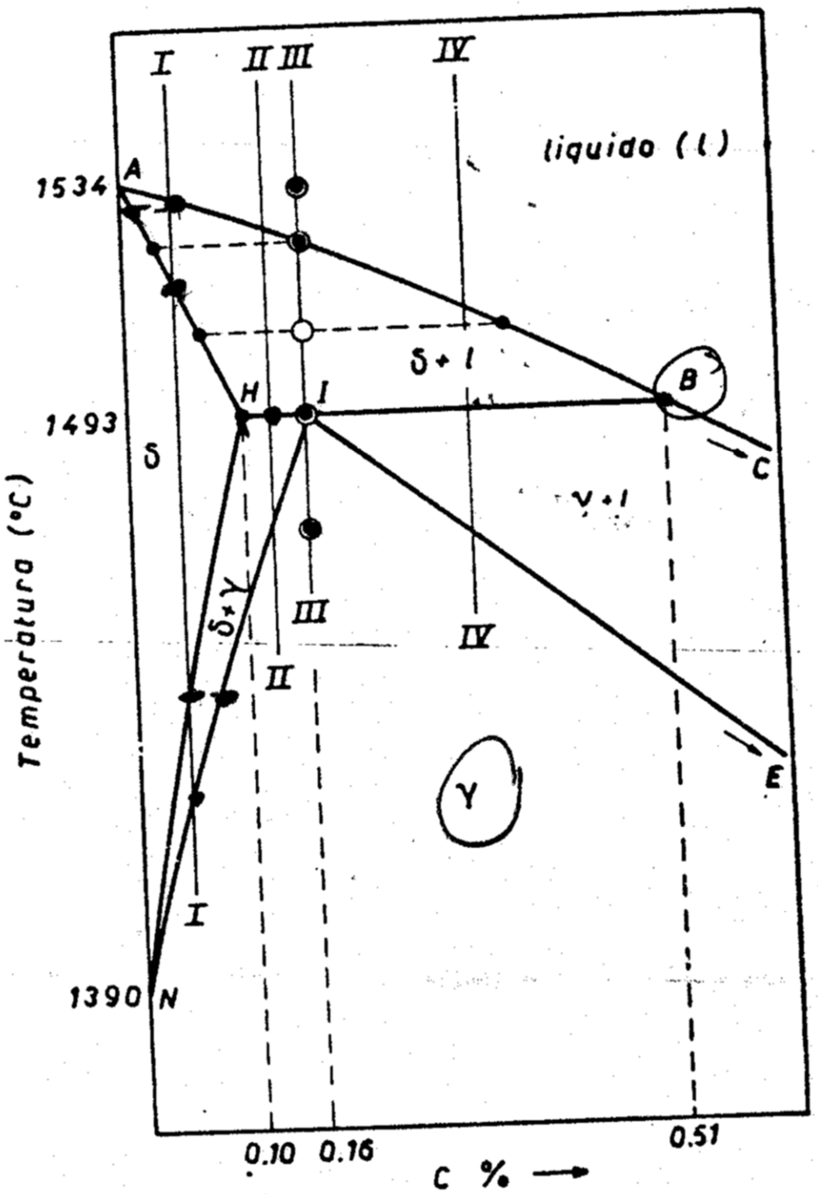



Fig. 3.51 - Trasformazione peritettica del diagramma Fe-C .

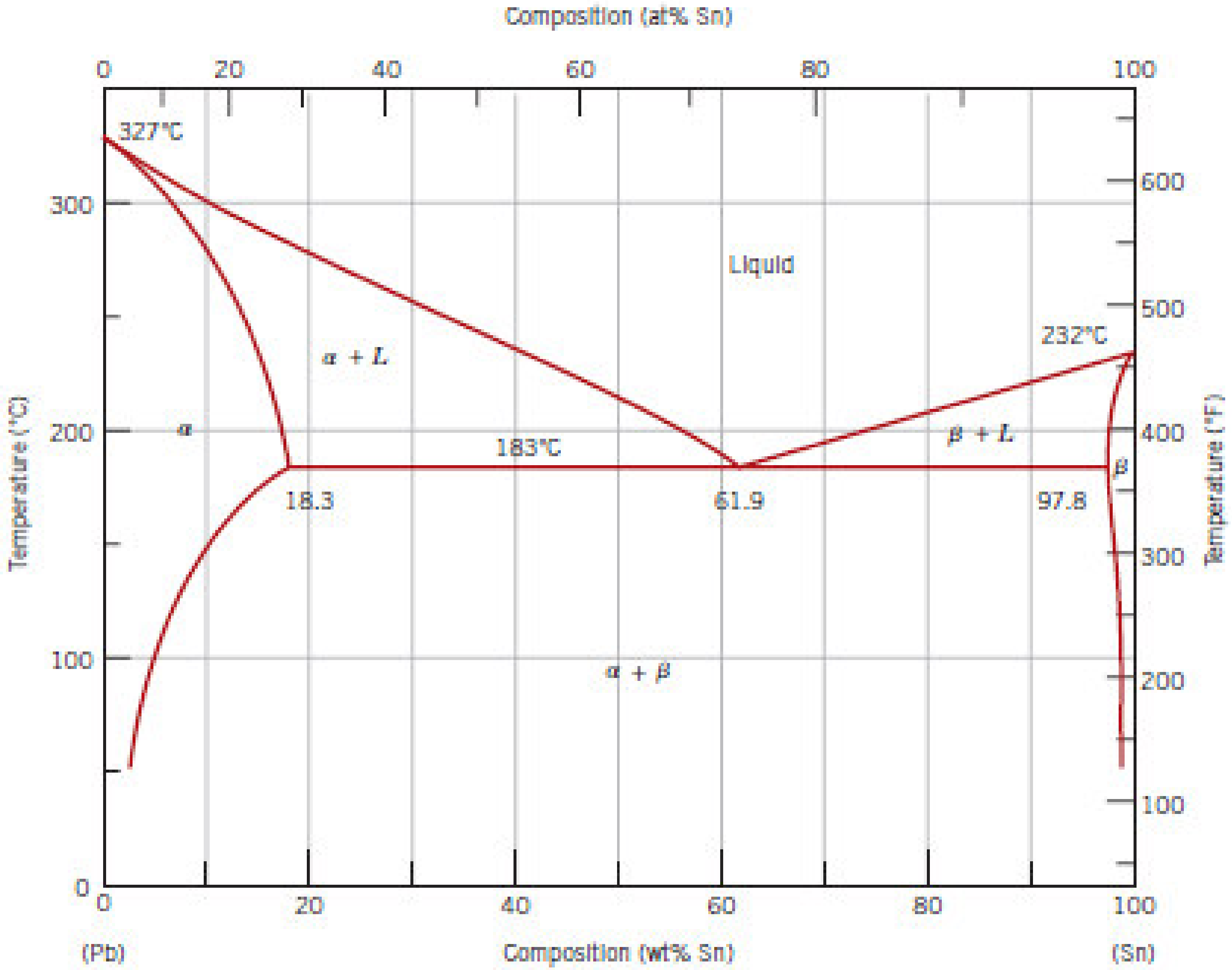



**Figure 9.8** The lead–tin phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 3, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

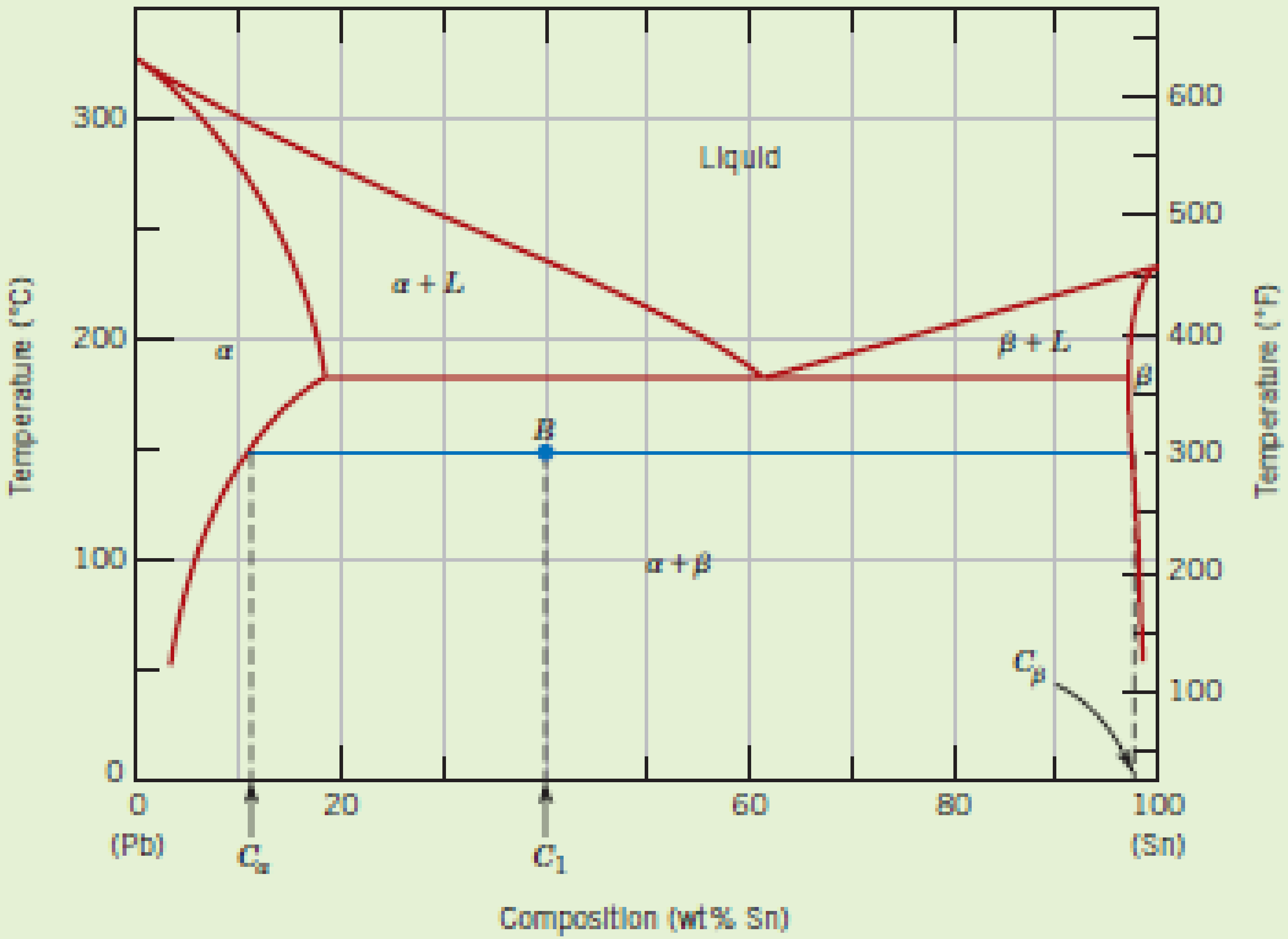



**Figure 9.9** The lead–tin phase diagram. For a 40 wt% Sn–60 wt% Pb alloy at 150°C (point $B$), phase compositions and relative amounts are computed in Example Problems 9.2 and 9.3.

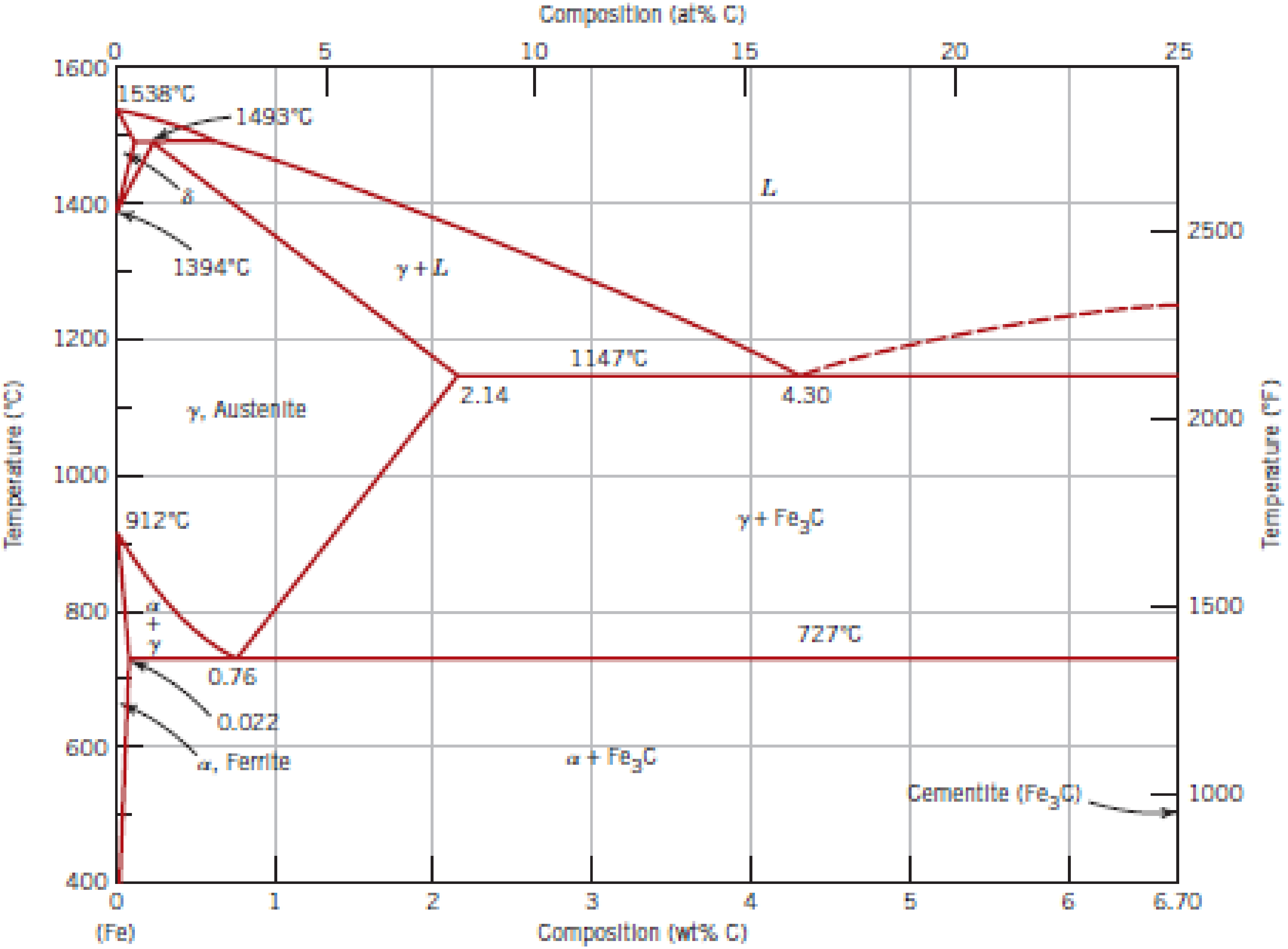



**Figure 9.24** The iron–iron carbide phase diagram. [Adapted from *Binary Alloy Phase Diagrams,* 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

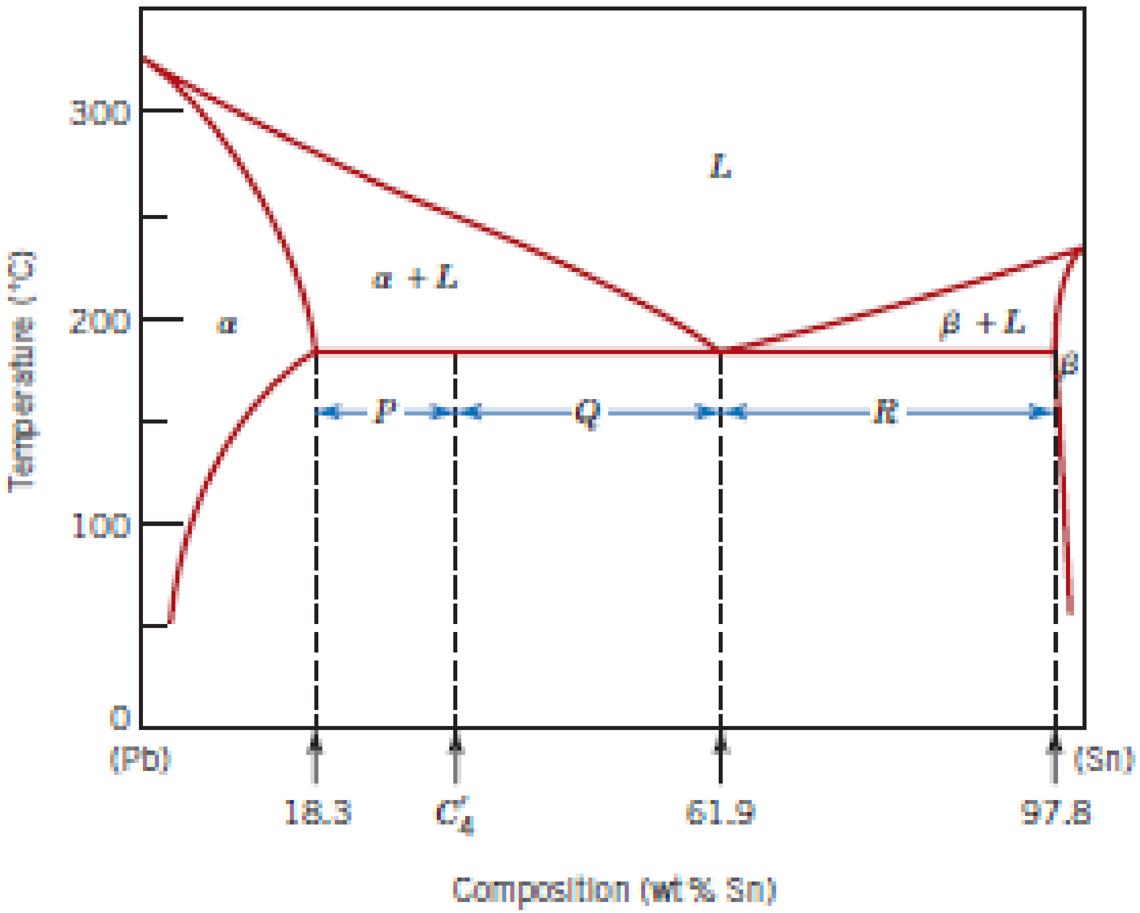



**Figure 9.18** The lead–tin phase diagram used in computations for relative amounts of primary $\alpha$ and eutectic microconstituents for an alloy of composition $C_4'$.

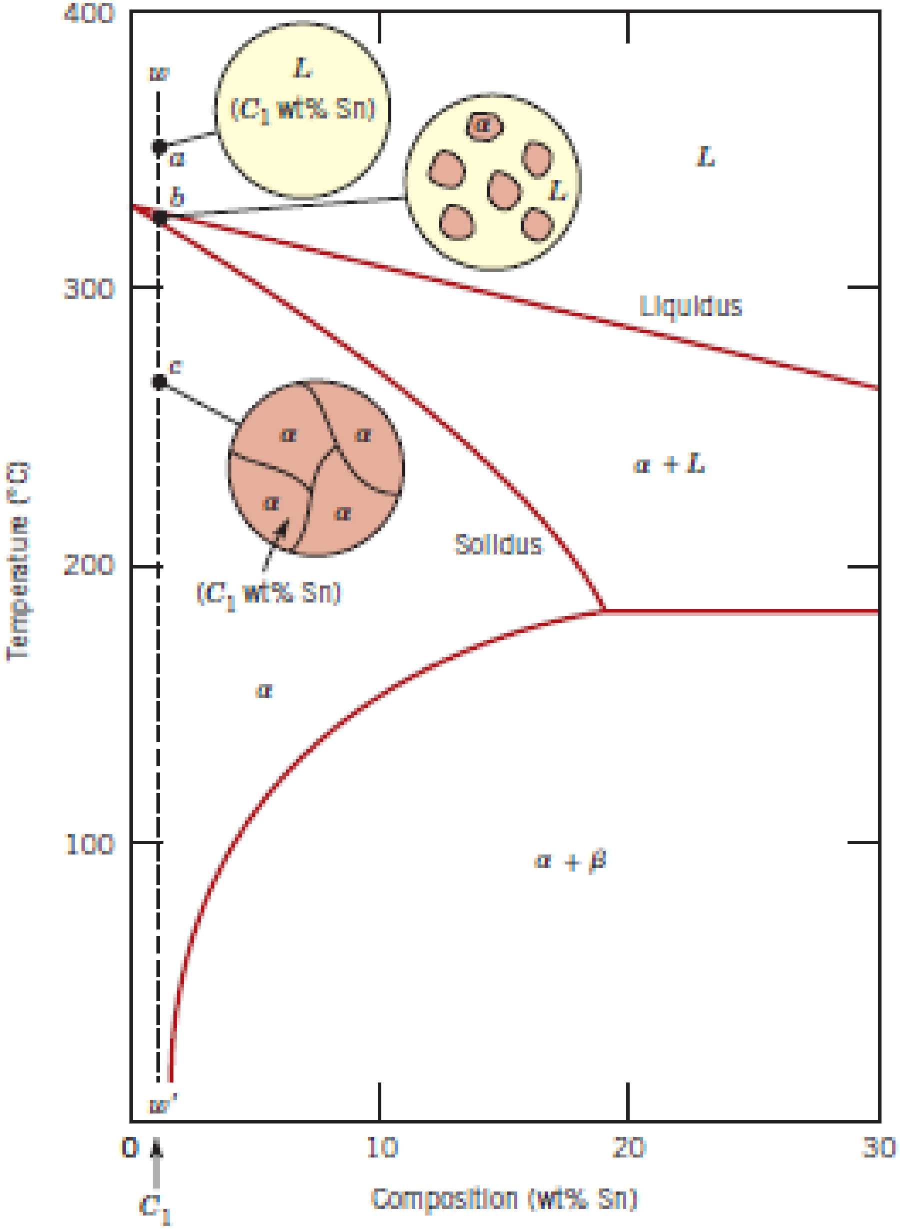



**Figure 9.11** Schematic representations of the equilibrium microstructures for a lead–tin alloy of composition $C_1$ as it is cooled from the liquid-phase region.

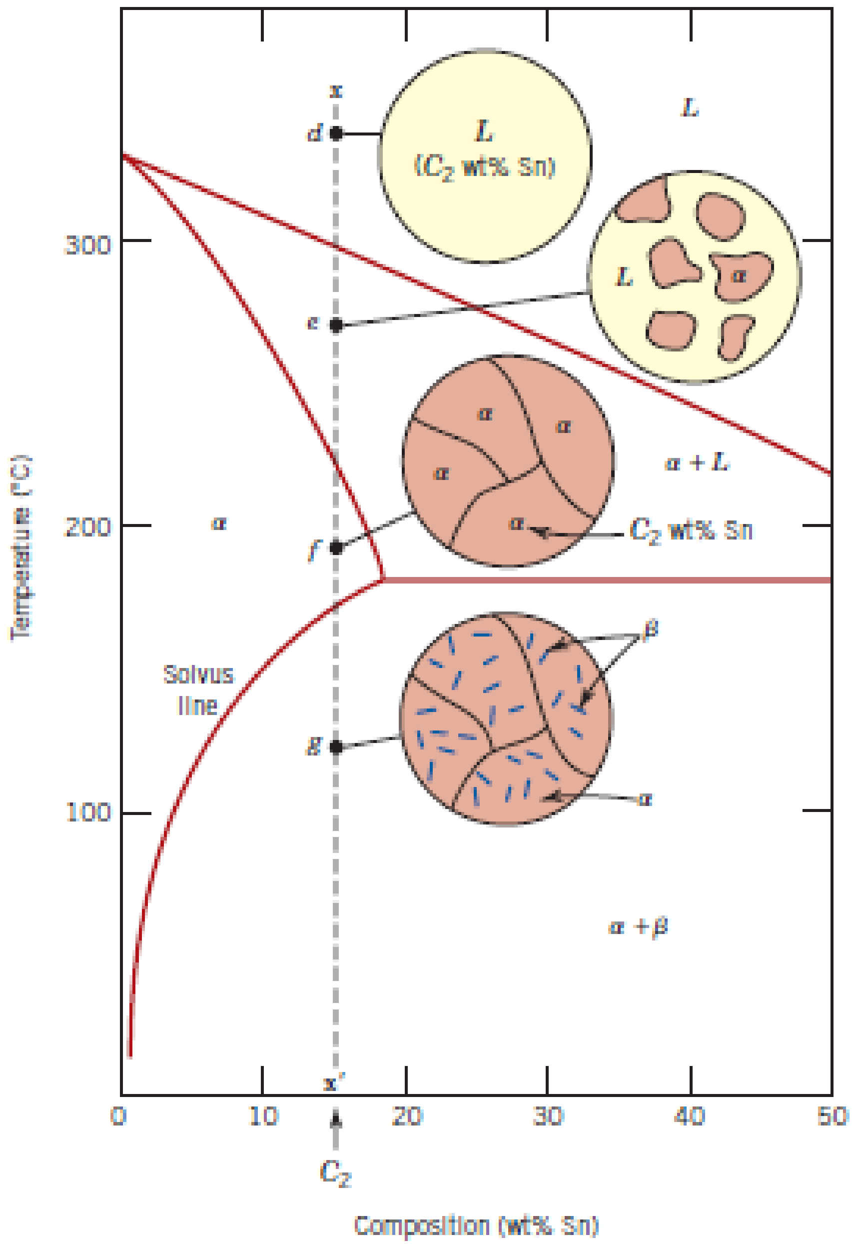



**Figure 9.12** Schematic representations of the equilibrium microstructures for a lead–tin alloy of composition $C_2$ as it is cooled from the liquid-phase region.

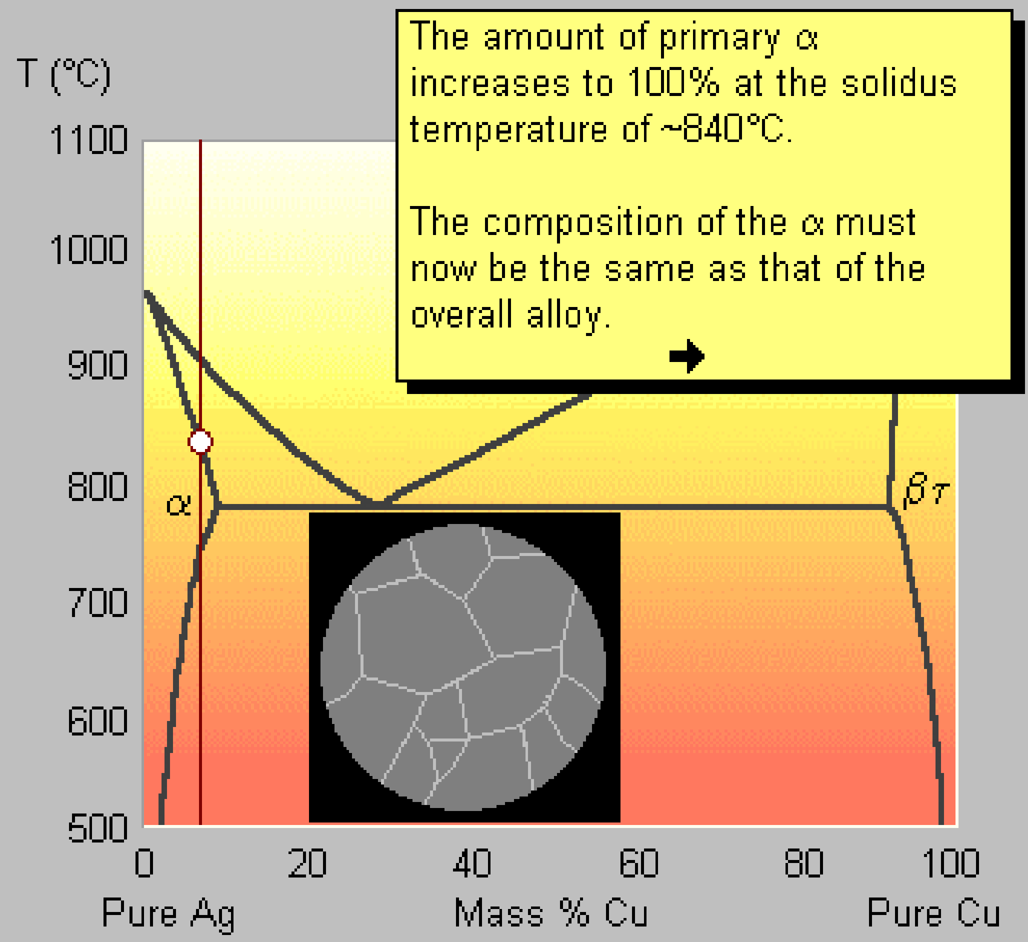
# Off-Eutectic Microstructure



- This alloy has a composition that lies outside the range of eutectic compositions.

Click on the button above the graph to start the solidification sequence.

After each stage of the animation, click on the ➜ symbol to proceed.



## Off-Eutectic Microstructure



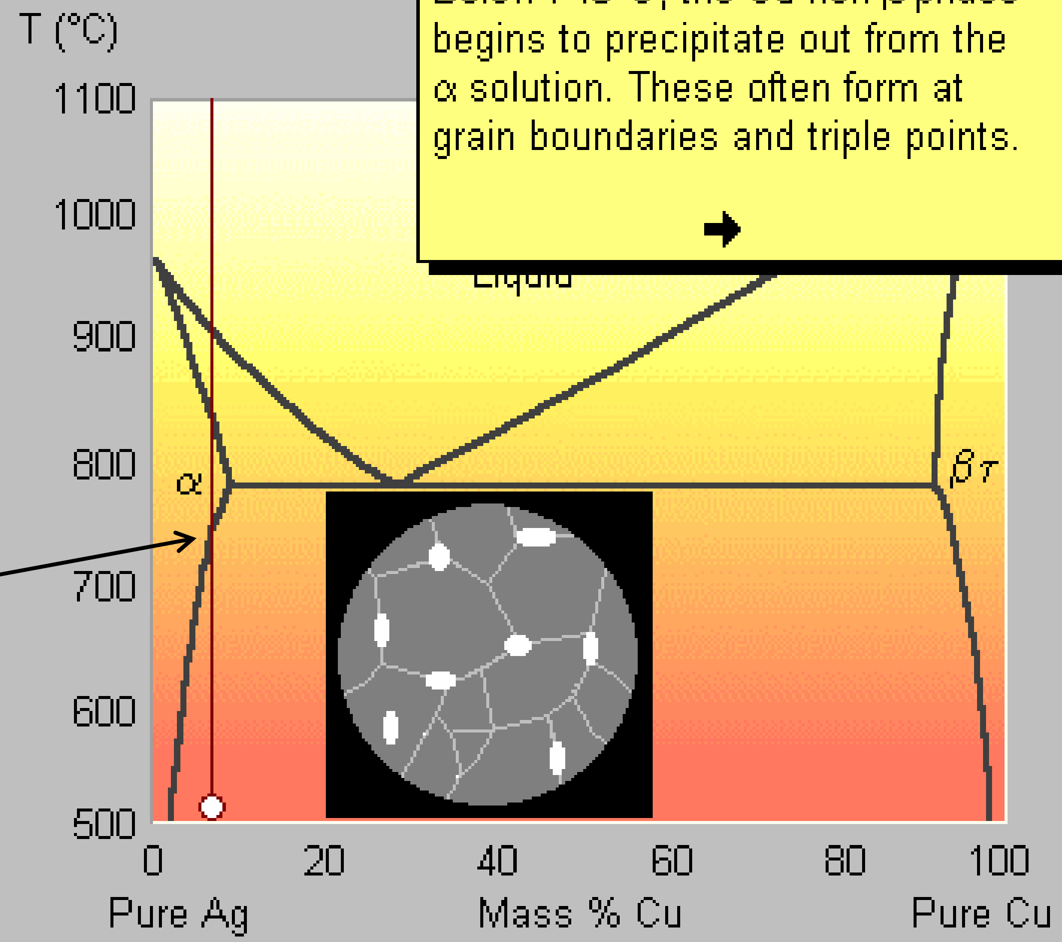
- This alloy has a composition that lies outside the range of eutectic compositions.

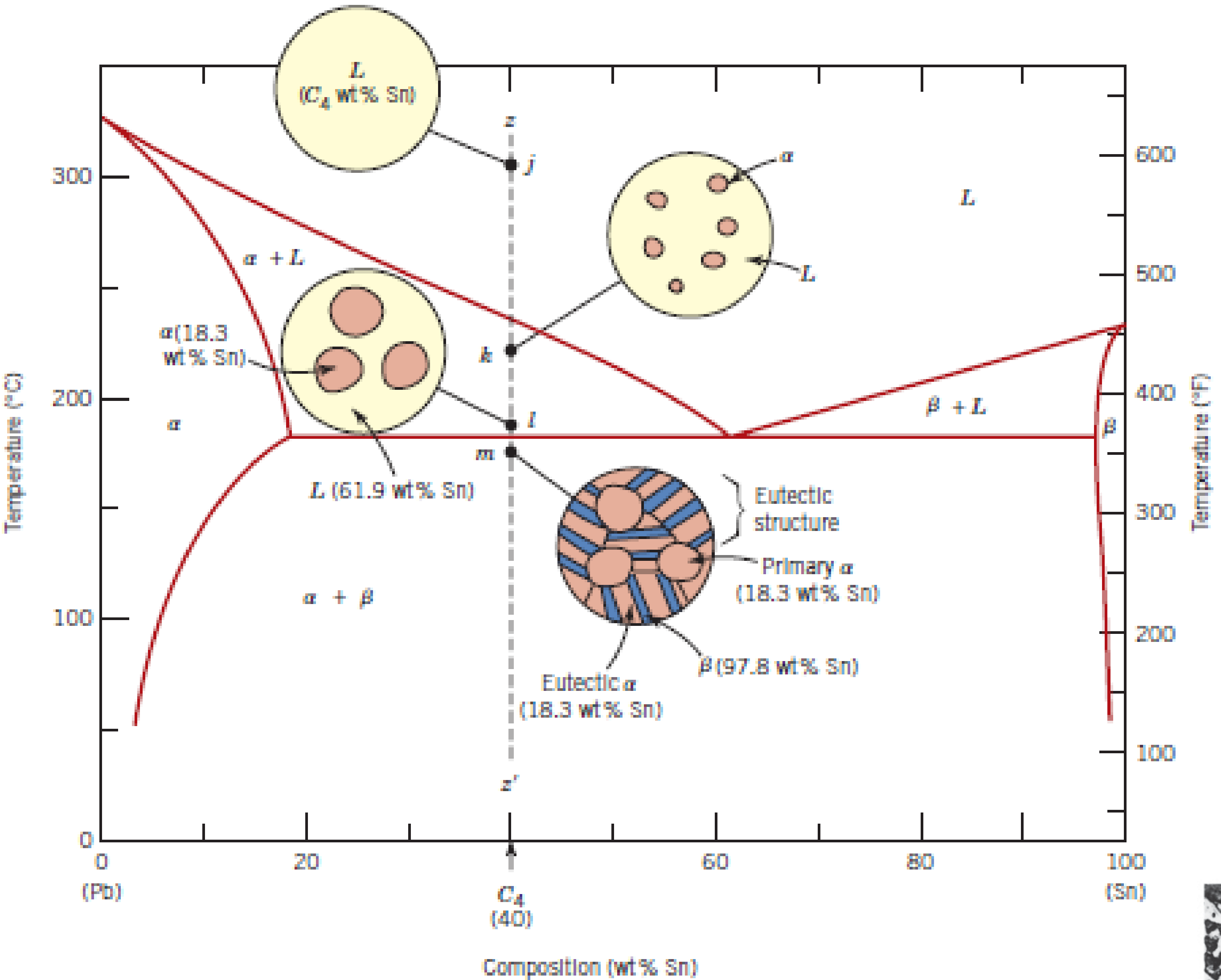
Click on the button above the graph to start the solidification sequence.

After each stage of the animation, click on the ➔ symbol to proceed.

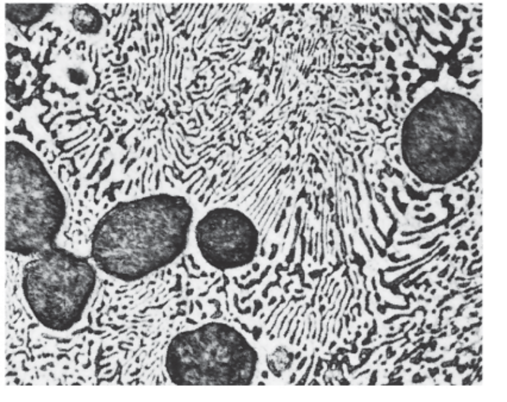
Supero il limite di solubilità:
Compare la seconda fase β

Below 740°C, the Cu-rich β phase begins to precipitate out from the α solution. These often form at grain boundaries and triple points.





L
($C_4$ wt% Sn)
z
j
α
L
α + L
k
α(18.3 wt% Sn)
l
α
m
L (61.9 wt% Sn)
Eutectic structure
Primary α (18.3 wt% Sn)
β + L
β
α + β
β (97.8 wt% Sn)
Eutectic α (18.3 wt% Sn)
z′
Temperature (°C)
Temperature (°F)
0
100
200
300
100
200
300
400
500
600
0
(Pb)
20
$C_4$
(40)
60
80
100
(Sn)
Composition (wt% Sn)

*l* and *m* are just above and below the euctectic T

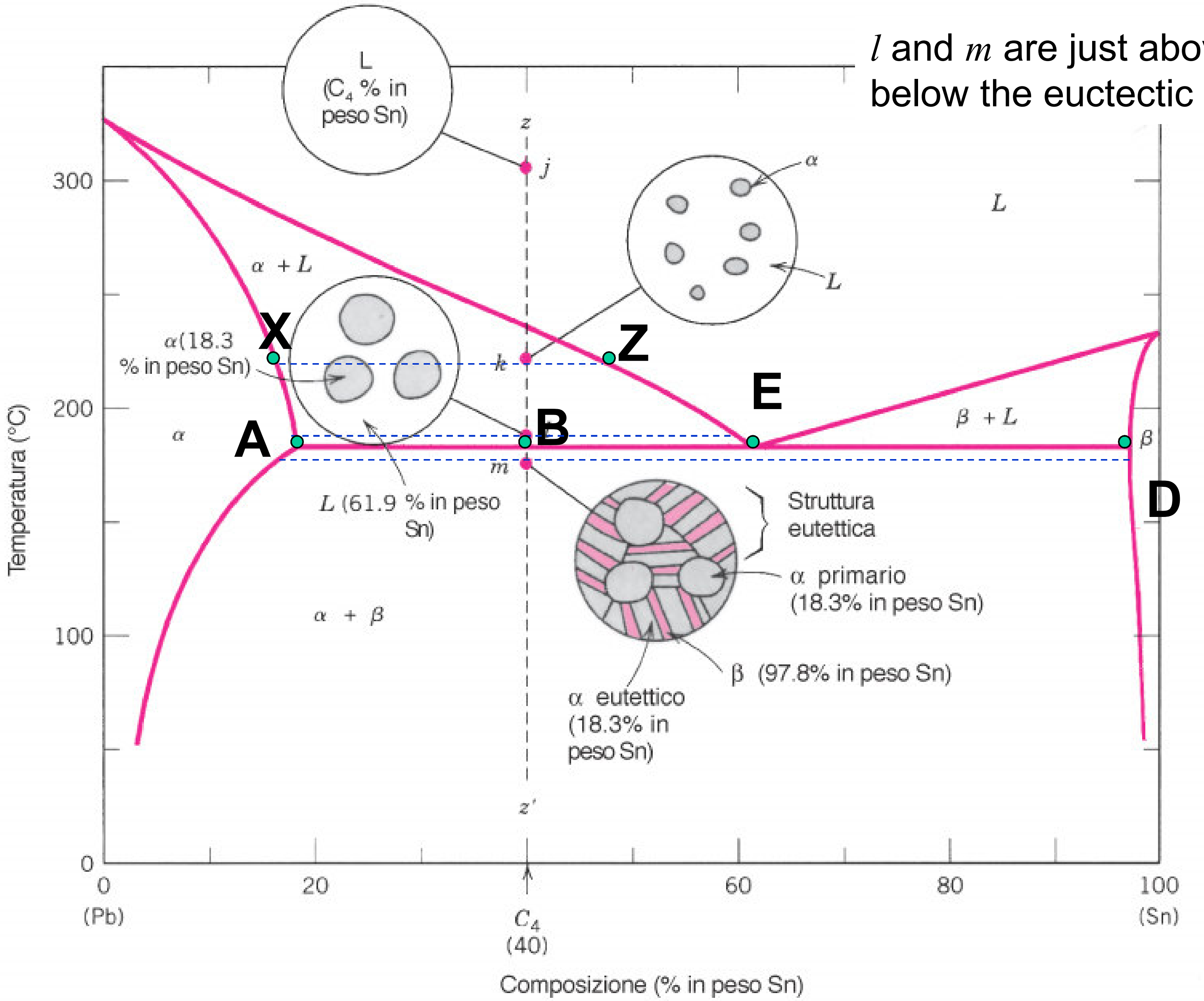



**FIGURA 9.14** Rappresentazione schematica delle microstrutture di equilibrio, assunte nel corso del raffreddamento a partire dal liquido dalla lega piombo–stagno di composizione $C_4$.

## Hypoeutectic Microstructure ( 1 )

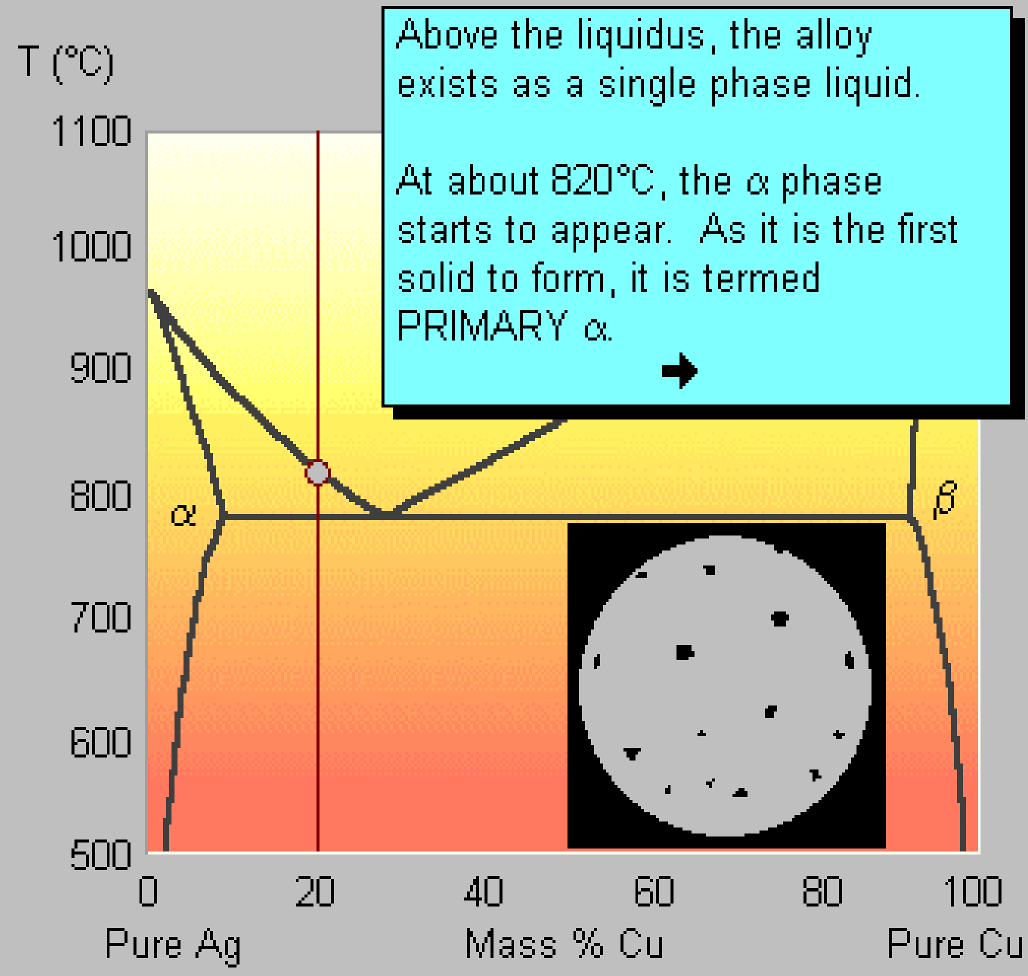


- In this alloy system, hypoeutectic alloys contain less Cu than the eutectic alloy.
- Here we shall look at the equilibrium cooling of a Ag-20%Cu alloy.

Start by clicking on the button above the graph.

Take time to compare the position of the grey temperature marker, the alloy microstructure and the textual descriptions.

After each stage of the animation, click on the ➡ symbol to proceed.



## Hypoeutectic Microstructure ( 1 )



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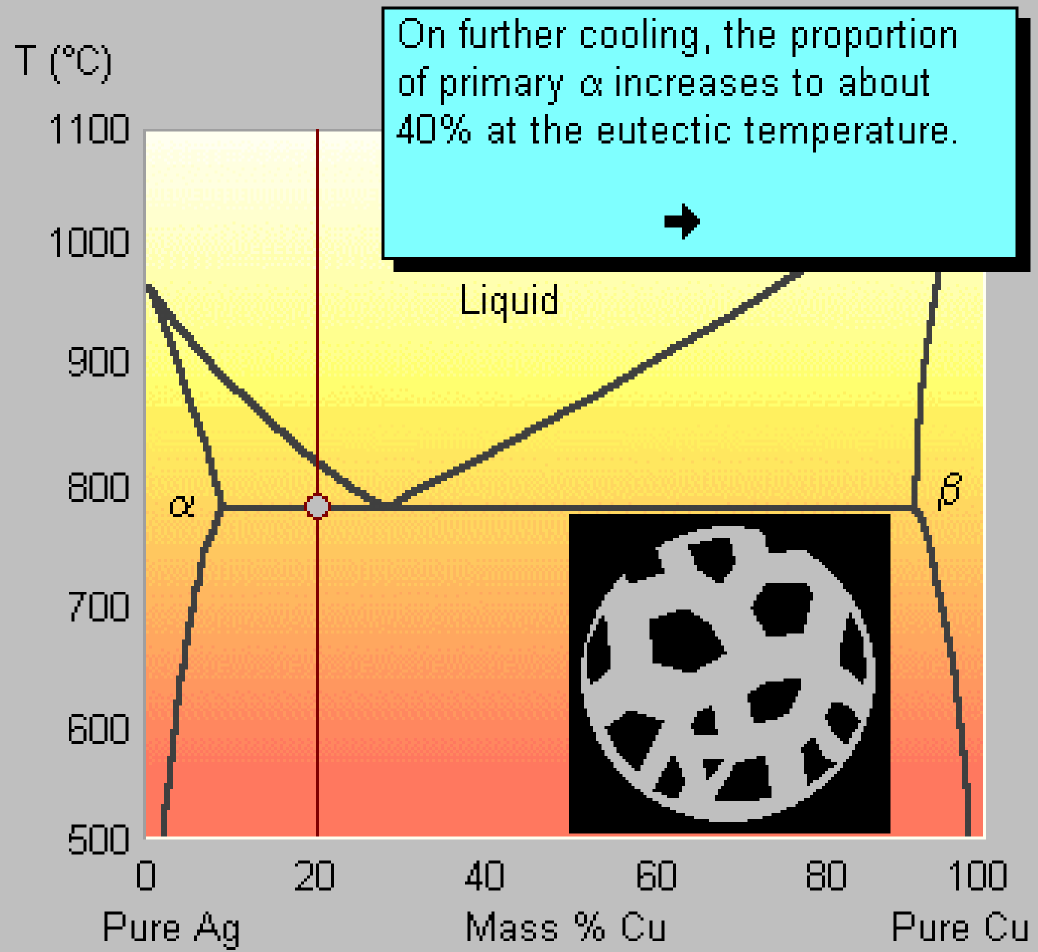
Start by clicking on the button above the graph.

Take time to compare the position of the grey temperature marker, the alloy microstructure and the textual descriptions.

After each stage of the animation, click on the ➔ symbol to proceed.



## Hypoeutectic Microstructure ( 2 )



- You can move the white constitution point to view the microstructure at any temperature.

  (NB The microstructure is only redrawn when you release the mouse button.)

- If you drag the mouse over different areas of the microstructure, the phase(s) are shown in the message box at the top of the page.

- Click here to view real micrograph of a Ag - 28% Cu alloy.

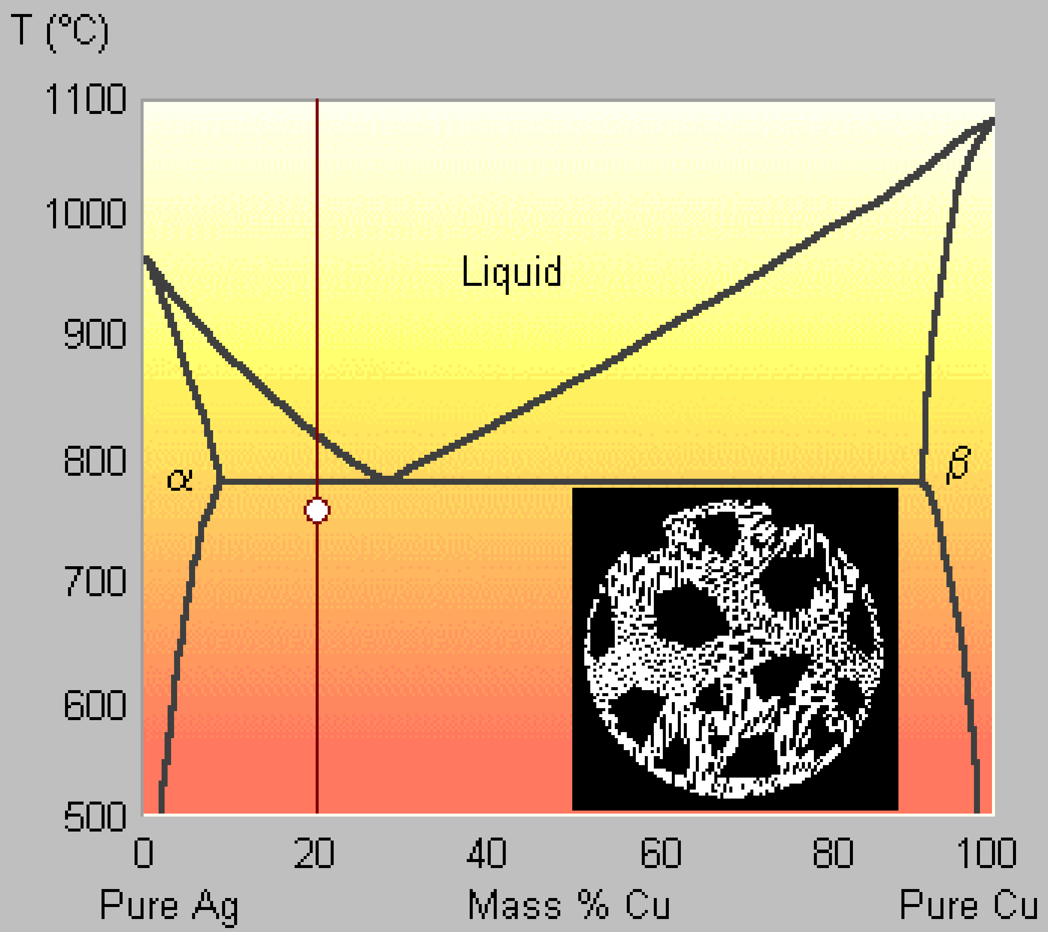
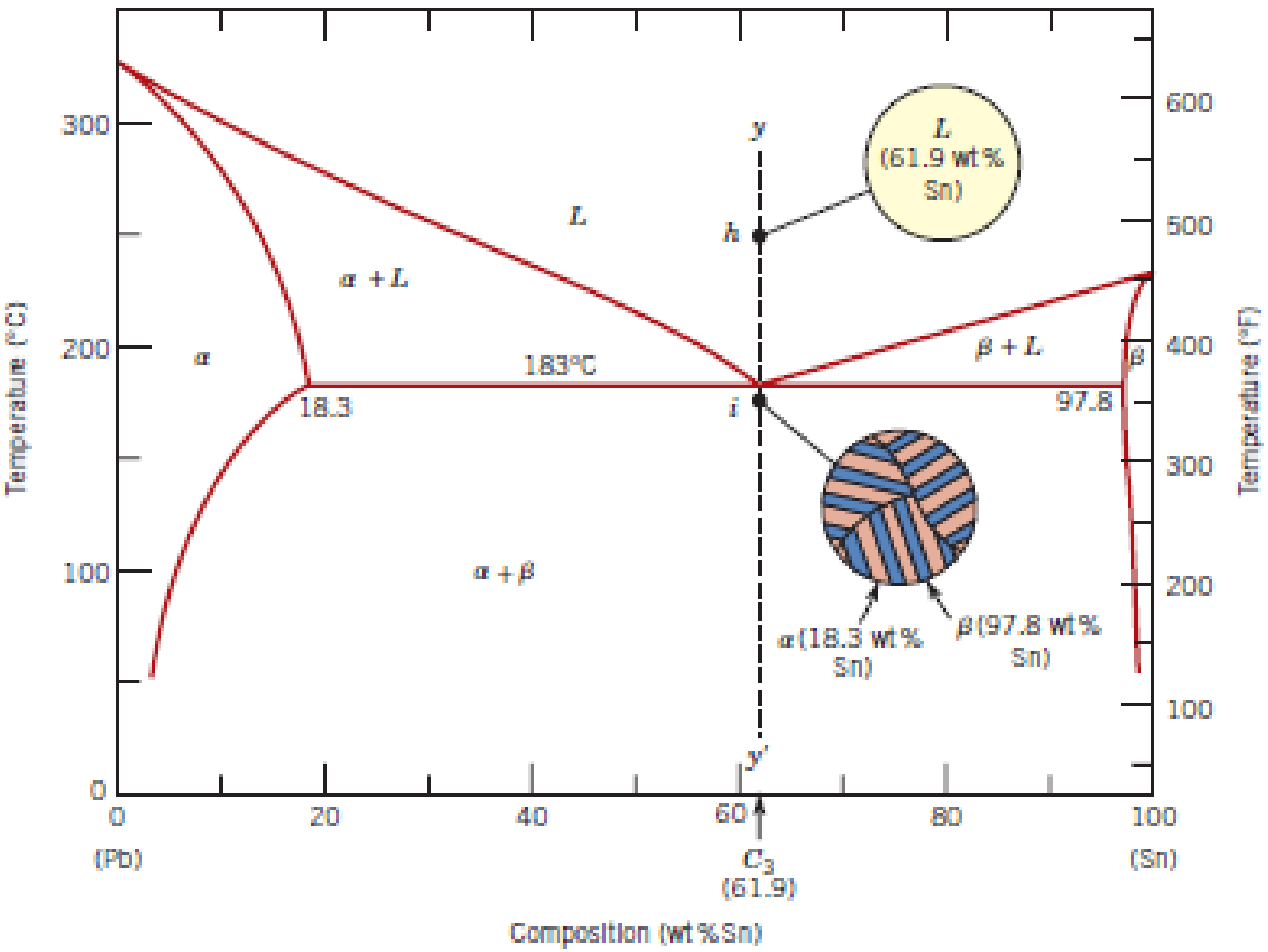


**Figure 9.13** Schematic representations of the equilibrium microstructures for a lead–tin alloy of eutectic composition $C_3$ above and below the eutectic temperature.



# Eutectic structure (Pb-Sn)

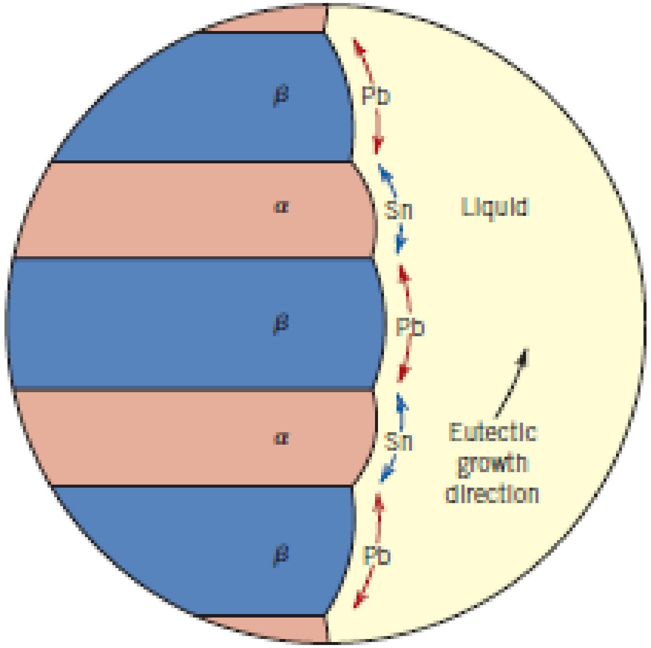



**Figure 9.15** Schematic representation of the formation of the eutectic structure for the lead–tin system. Directions of diffusion of tin and lead atoms are indicated by blue and red arrows, respectively.

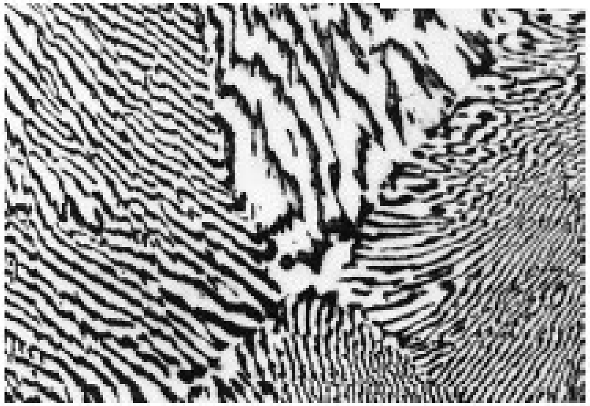


**Figure 9.14** Photomicrograph showing the microstructure of a lead–tin alloy of eutectic composition. This microstructure consists of alternating layers of a lead-rich α-phase solid solution (dark layers), and a tin-rich β-phase solid solution (light layers). 375×. (Reproduced with permission from *Metals Handbook*, 9th edition, Vol. 9, *Metallography and Microstructures*, American Society for Metals, Materials Park, OH, 1985.)

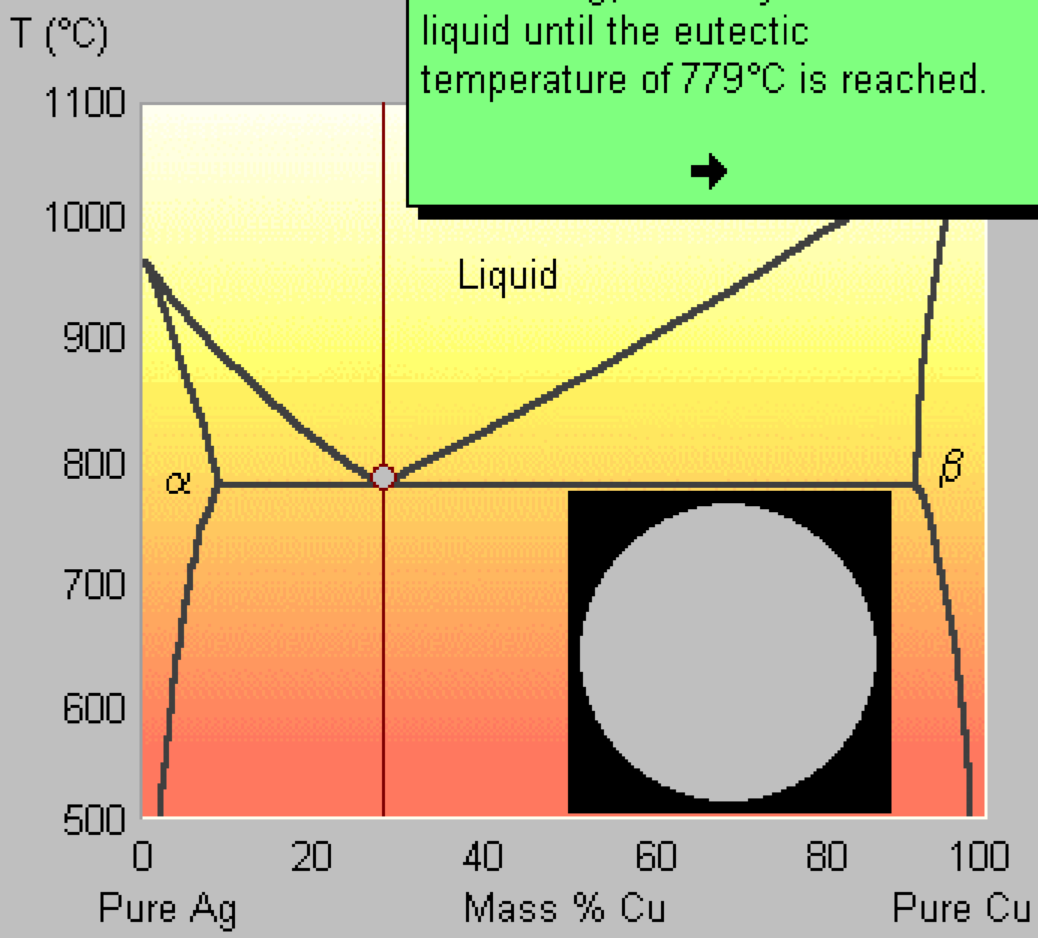
# RAFFREDDAMENTO DI LEGHE EUTETTICHE 2

EUTECTIC ALLOYS **Eutectic Microstructure ( 1 )** 10 of 16

- We shall first look at the development of microstructure for the alloy of eutectic composition (29%Cu).
- Start by clicking on the button above the graph.
- After each stage of the animation, hit the ➜ symbol to proceed.

On cooling, the alloy remains liquid until the eutectic temperature of 779°C is reached.

➜



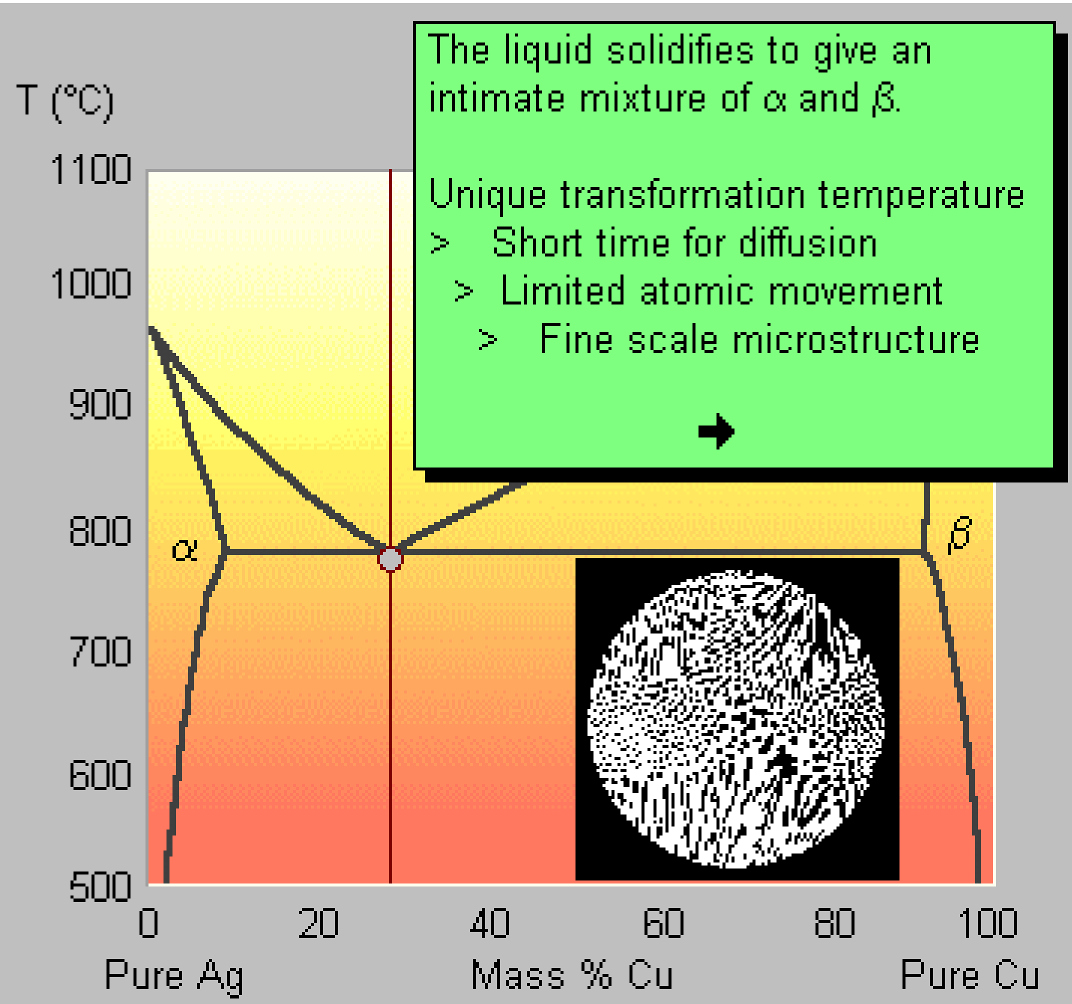
# RAFFREDDAMENTO DI LEGHE EUTETTICHE 3

EUTECTIC ALLOYS

## Eutectic Microstructure ( 1 )



- We shall first look at the development of microstructure for the alloy of eutectic composition (29%Cu).
- Start by clicking on the button above the graph.
- After each stage of the animation, hit the ➜ symbol to proceed.

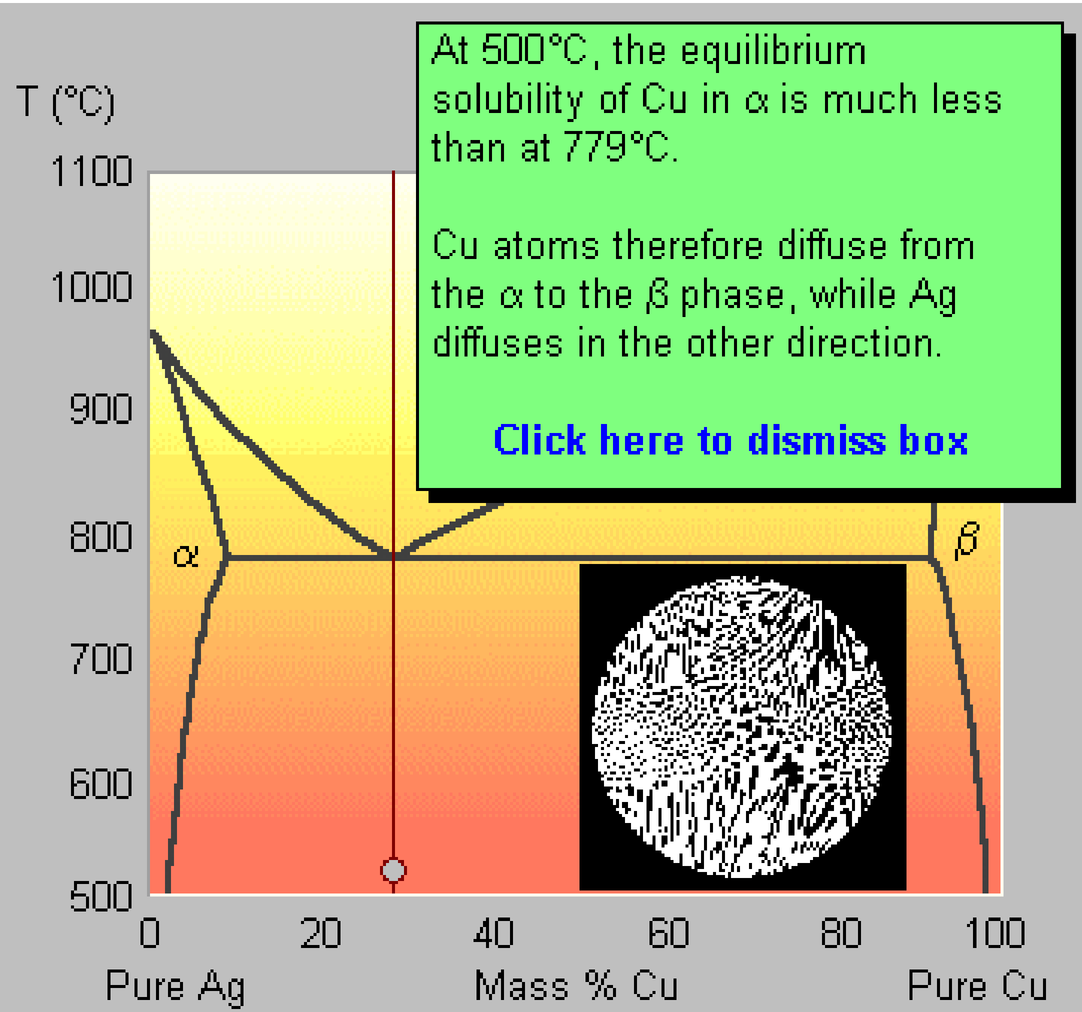


# RAFFREDDAMENTO DI LEGHE EUTETTICHE 3



- We shall first look at the development of microstructure for the alloy of eutectic composition (29%Cu).

Abbassando la Temperatura al di sotto di $T_E$ le fasi restano alfa e beta, ma cambiano di composizione e in rapporto quantitativo seguendo le curve di solvus



# RAFFREDDAMENTO DI LEGHE IPEREUTETTICHE



## Hypereutectic Microstructure ( 1 )



- Hypereutectic alloys contain more Cu than the eutectic alloy.
- In this example we shall look at the equilibrium cooling of a Ag-65%Cu alloy.

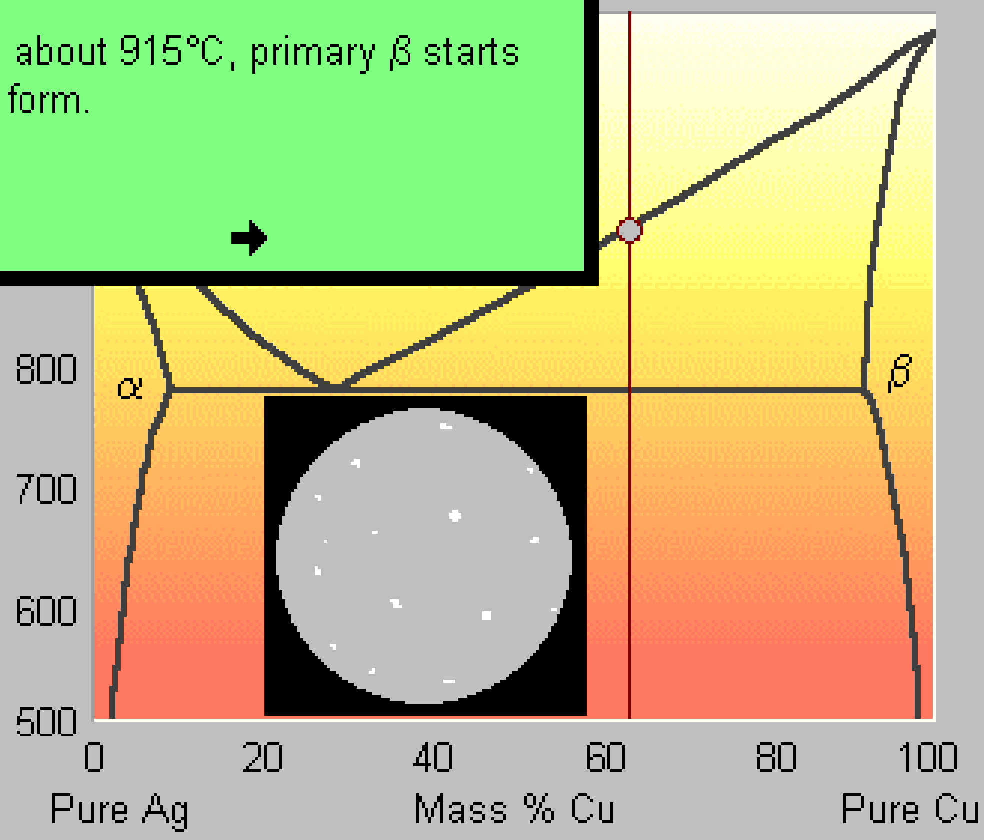
Click on the button above the graph to start the solidification sequence.

After each stage of the animation, hit the ➡ symbol to proceed.

Above the liquidus, the alloy exists as a single phase liquid.

At about 915°C, primary β starts to form.

➡



# RAFFREDDAMENTO DI LEGHE IPEREUTETTICHE 2



**Hypereutectic Microstructure ( 1 )**

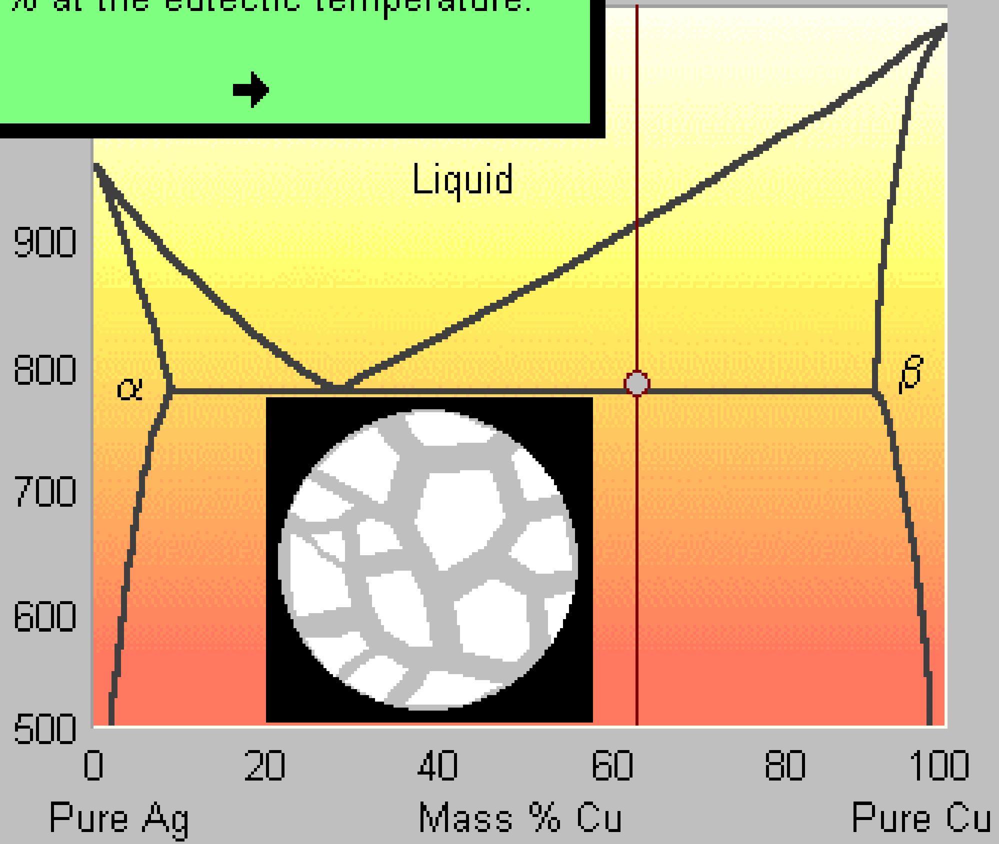


- Hypereutectic alloys contain more Cu than the eutectic alloy.
- In this example we shall look at the equilibrium cooling of a Ag-65%Cu alloy.

Click on the button above the graph to start the solidification sequence.

After each stage of the animation, hit the ➜ symbol to proceed.

On further cooling, the proportion of primary β increases to about 55% at the eutectic temperature.



# RAFFREDDAMENTO DI LEGHE IPEREUTETTICHE 3



**Hypereutectic Microstructure ( 1 )**



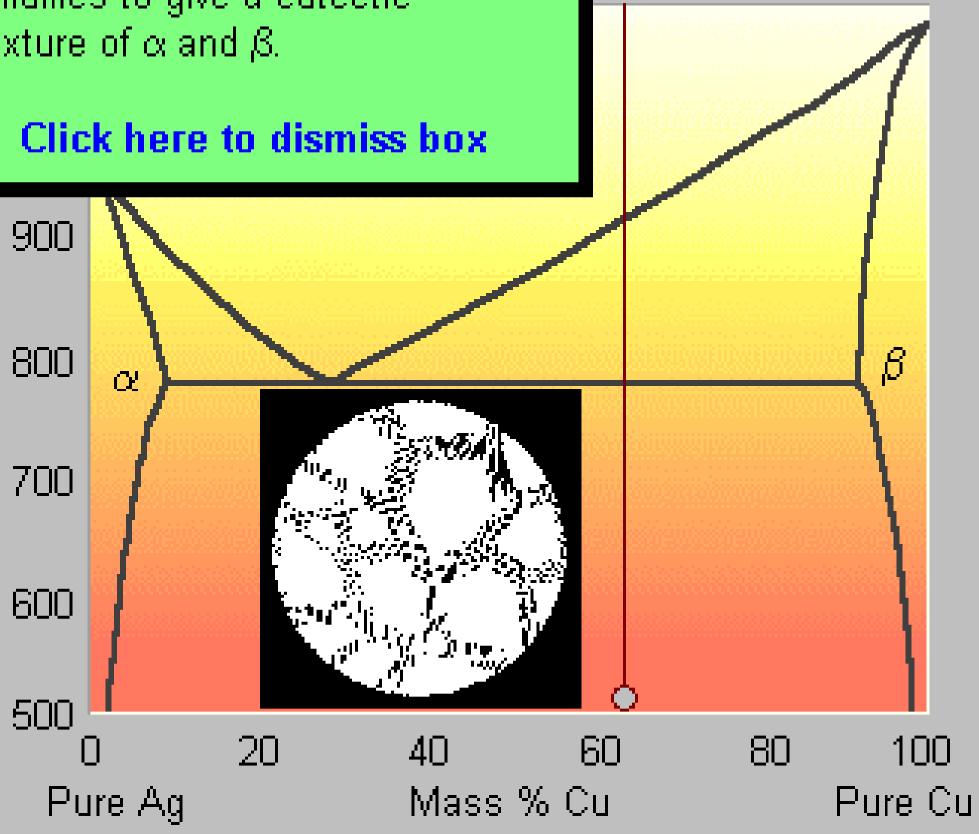
- Hypereutectic alloys contain more Cu than the eutectic alloy.
- In this example we shall look at the equilibrium cooling of a Ag-65%Cu alloy.

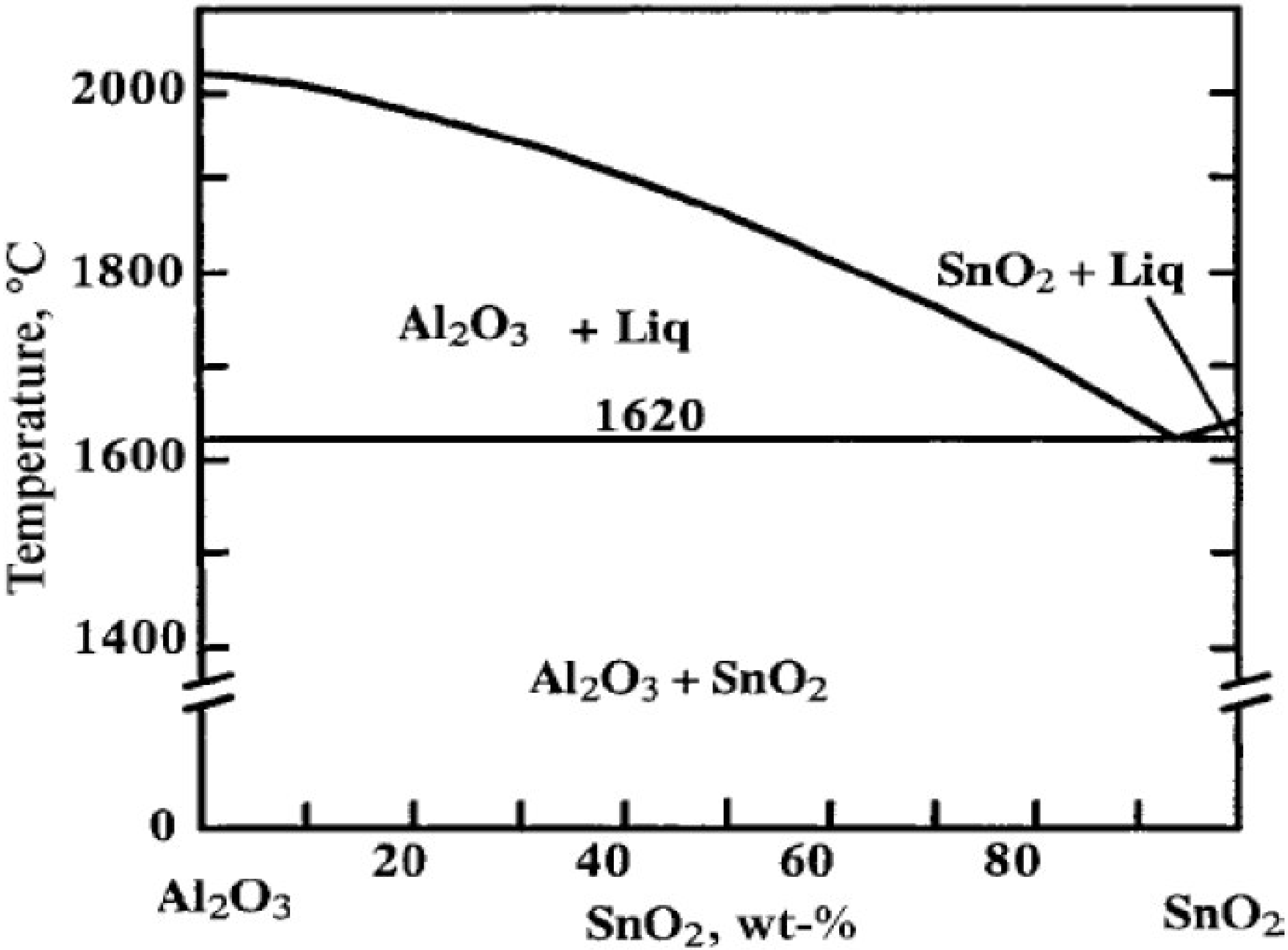
Click on the button above the graph to start the solidification sequence.

After each stage of the animation, hit the ➔ symbol to proceed.

As the eutectic temperature is passed, the remaining liquid solidifies to give a eutectic mixture of α and β.

**Click here to dismiss box**





2000
1800
1600
1400
0
Temperature, °C
Al2O3 + Liq
SnO2 + Liq
1620
Al2O3 + SnO2
20
40
60
80
Al2O3
SnO2, wt-%
SnO2

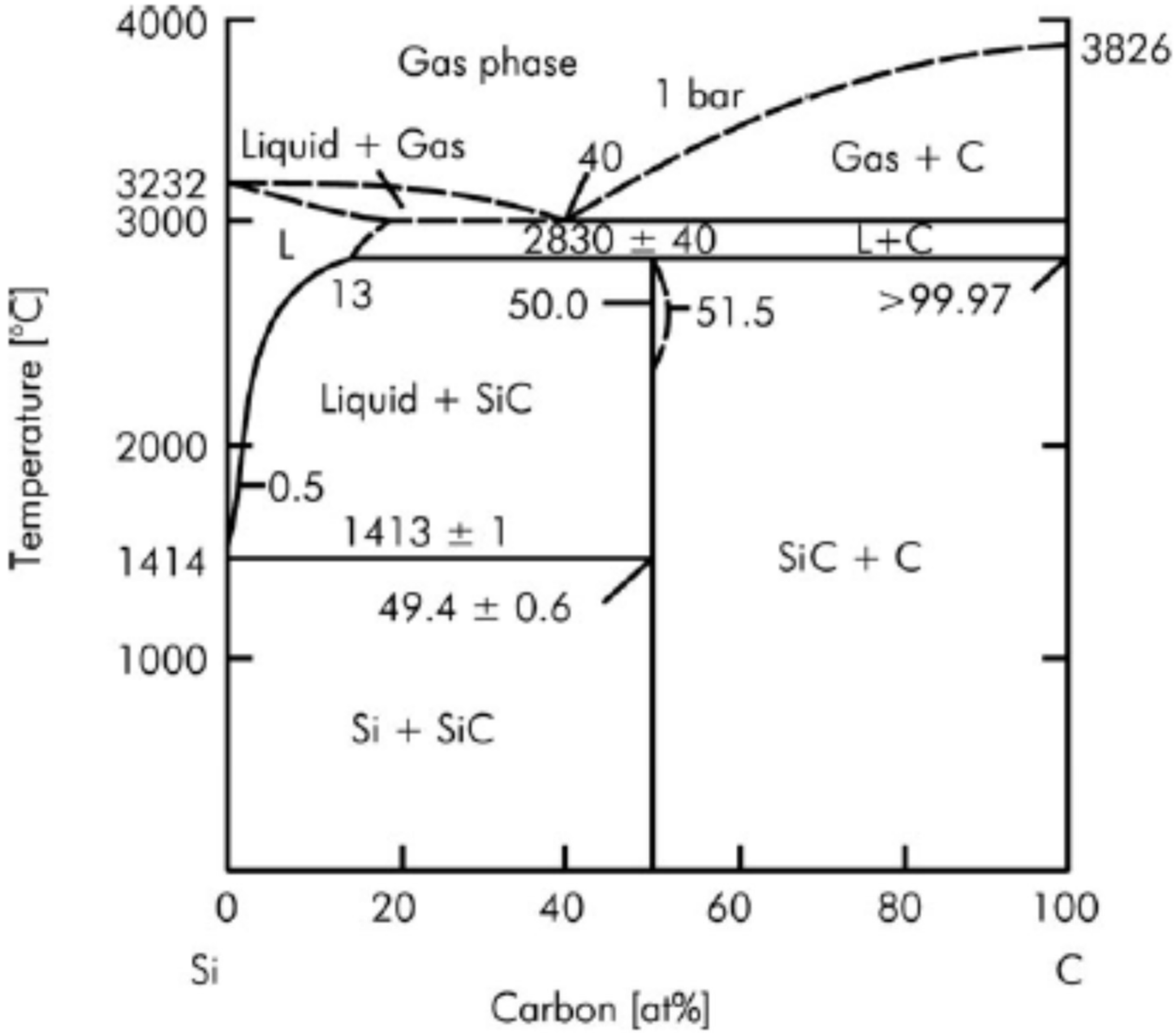
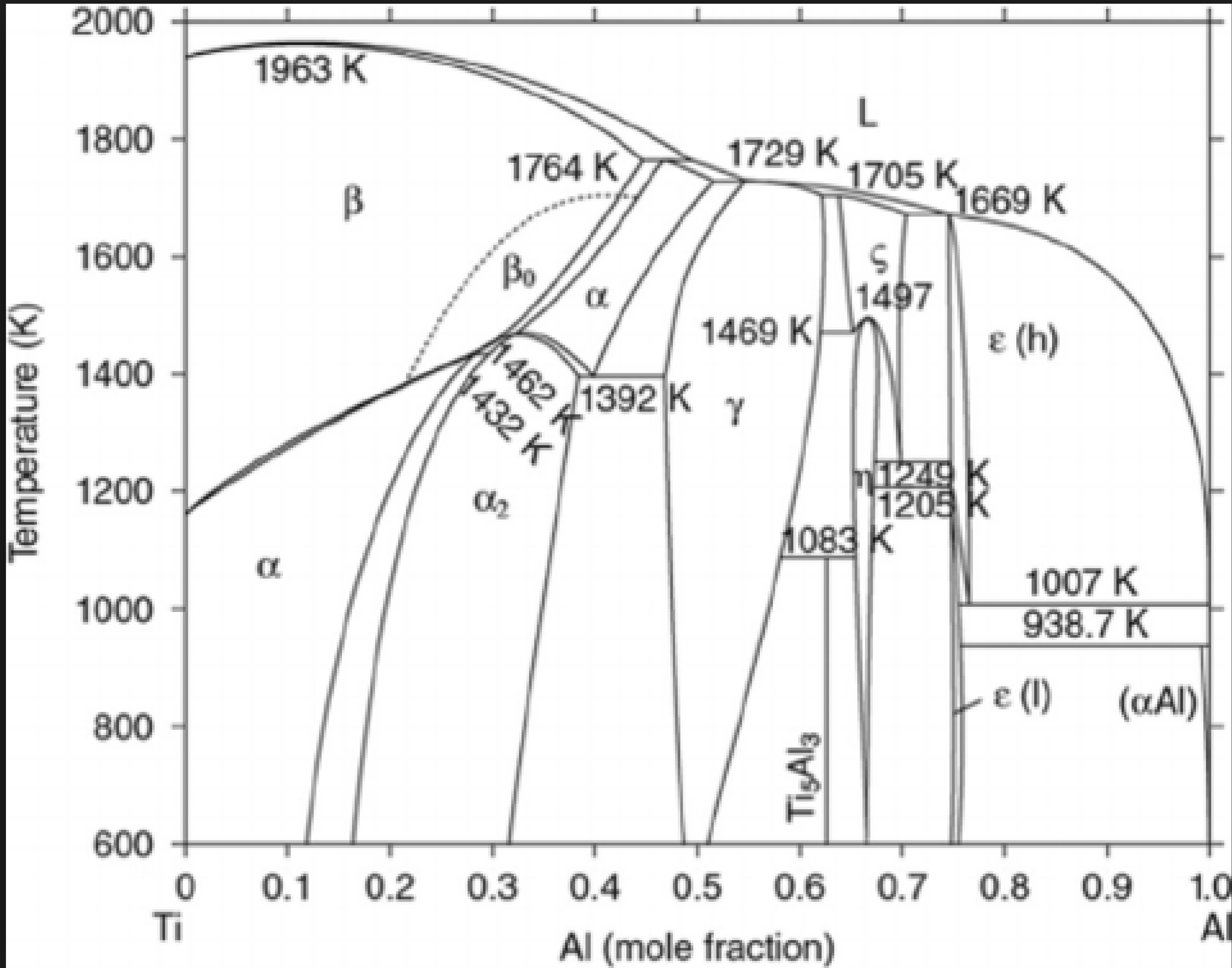



**FIGURE 12** Si-C phase diagram from[24]



2000
1800
1600
1400
1200
1000
800
600
Temperature (K)
1963 K
L
1764 K
1729 K
1705 K
1669 K
β
$\beta_0$
α
ς
1497
1469 K
ε (h)
1462 K
1432 K
1392 K
γ
η
1249 K
1205 K
$\alpha_2$
1083 K
α
1007 K
938.7 K
ε (l)
(αAl)
$Ti_5Al_3$
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Ti
Al
Al (mole fraction)